

Generative AI and the New Landscape of Automated Journalism: A Systematized Review of 185 Studies (2012–2024)

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Abstract

The rapid acceleration of artificial intelligence (AI) and, more recently, generative AI is reshaping journalism in ways that extend far beyond earlier forms of news automation. As generative AI tools become widely accessible and capable of processing unstructured data, longstanding definitions of automated journalism—once centered on structured datasets and template-based text generation—are being fundamentally reconfigured. This paper presents the most comprehensive and up-to-date systematized review of automated journalism scholarship, expanding on earlier research by synthesizing 185 peer-reviewed, English studies published between 2012 and 2024 about machine-generated textual news content published online. Through a rigorously designed search strategy across four major social science databases, this review maps how the field’s conceptual, methodological, and geographical contours have transformed as AI and generative AI become increasingly ubiquitous. The findings show a surge of research in 2024 alone, as well as the emergence of more than 150 overlapping terms used to describe AI- and algorithmically generated news, illustrating significant conceptual fragmentation. Despite no overly dominant theories, concepts or frameworks, key themes include credibility and trust, human–machine collaboration, newsroom adoption and institutional logics, transparency and disclosure, and the ethical and regulatory challenges introduced by increasingly sophisticated AI systems. By consolidating patterns, evaluating an expanded selection of key terms, and assessing theoretical and conceptual frameworks, this review demonstrates how AI and especially generative AI reflect the speed of industrial change, but also the lack of shared academic frameworks to make sense of that change.

Keywords: automated journalism; artificial intelligence (AI); generative AI; algorithmic news production; computational journalism; systematized review; news automation; human–machine communication



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1. Introduction

Thanks to increasingly abundant and openly available data, combined with artificial intelligence (AI)-enabled tools, over the past decade there has been a steady increase in news outlets publishing machine-generated textual content, known broadly as automated journalism ([Diakopoulos, 2019](#)). However, with the recent rise of generative AI, which began in 2014 with structured earthquake data being automatically fed into a pre-written template and sent to the Los Angeles Times’ copy desk for proofing ([Oremus, 2014](#)), this has exploded into a myriad of possibilities. Large language models can validate subjective game

insights of sports journalists (Cheng et al., 2024) or produce real-time match summaries from video (Sarkar et al., 2024). Machine learning can generate complete news articles from geo-tagged images (Gotmare et al., 2024) or the transcripts of legislative proceedings (Klimashevskaja et al., 2021). Text can be quickly edited, translated, and formatted to align with the style and tone of news outlets (Newman, 2024). This fundamental shift in what constitutes automated journalism in practice demands a renewed exploration into what falls under the same banner within scholarship.

While automated journalism has, to date, been most understood to represent algorithmically generated news texts, a number of terms have been used interchangeably. Initially called robot journalism (or robo-journalism) (Carlson, 2015; Clerwall, 2014; Latar, 2018), this term has generally fallen out of favor because such content production does not actually involve robotics (Danzon-Chambaud, 2021). Algorithmic journalism (Dörr, 2016) includes machine-generated content but can also be used to describe several other algorithmic applications in journalism, like data mining, content optimization, and news dissemination (Kotenidis & Veglis, 2021). Since automation can increasingly be used to generate other types of content, such as images or video, or could be used to describe the automated publication of news rather than its production (Danzon-Chambaud, 2021; S. Wu et al., 2019), it has become increasingly common to use more specific terms to describe machine-generated news texts, including computer-generated news (Graefe et al., 2016), automated content production (Kotenidis & Veglis, 2021), auto-written news stories (Y. Wu, 2020), automated news/news automation (Dierickx, 2023; Haim & Graefe, 2017), and AI-generated news (S. Kim & Kim, 2020). When we began this study, we had already documented 25 different terms being used to describe automated journalism. By the end, that list had expanded to more than 150.

With the widespread and rapid uptake of easy-to-use generative AI technologies, the amount of research on automated journalism has also grown over the past decade (Danzon-Chambaud, 2021), and this trend is likely to accelerate. Studies to date have explored audience perceptions of automated news content (Graefe et al., 2016; Graefe & Bohlen, 2020; Jung et al., 2017; Y. Kim & Lee, 2021; Noain-Sánchez, 2022; Waddell, 2019b; Y. Wu, 2020); the impact on jobs, news organizations, and newsroom routines (Danzon-Chambaud & Cornia, 2021; Dierickx, 2023; Wagner, 2022); questions about journalistic authority and authorship (Carlson, 2015; Schapals & Porlezza, 2020); ethical implications (Dörr & Hollnbuchner, 2017; Rojas Torrijos, 2021); and automating more complex news stories (Caswell & Dörr, 2018, 2019).

While there is a fear that such automated content will further decrease the availability of jobs in an already decimated industry (St-Germain & White, 2021), research suggests it in fact supplements the work of journalists (Carlson, 2015; Clerwall, 2014; Diakopoulos, 2019; Schapals, 2020; Noain-Sánchez, 2022). Automated journalism has the potential to improve the quality of news content and news by providing audiences with more relevant information (Peiser, 2019; Wagner, 2022), increasing the credibility of news content (Graefe et al., 2016; Waddell, 2018, 2019a; Y. Wu, 2020), reducing financial burdens on news organizations (Wagner, 2022), and freeing up journalists' time, efforts and energy for more impactful work (Clerwall, 2014; Diakopoulos, 2019; Schapals, 2020). With the uptake of generative AI, automated journalism has the further potential to leverage unstructured data, such as quotes, transcripts, and images, to create richer content for news audiences that moves beyond repetitive and dry pre-formatted templates without further taxing the resources of journalists and news organizations (Caswell & Dörr, 2018, 2019; Cheng et al., 2024; Sarkar et al., 2024). Techno-optimists argue that these powerful tools will serve to revitalize journalism by expanding journalists' capabilities, engaging fragmented news audiences, and making content more global and more accessible (Munoriyarwa & de-Lima-

Santos, 2025). However, generative AI presents challenges for journalism that go beyond questions of job replacement, content quality, transparency and authorship, and instead strike at the heart of the concepts of journalistic agency, power, and authority (Munoriyarwa & de-Lima-Santos, 2025). To that end, it is imperative that scholars better understand what constitutes automated journalism, how it is being deployed, and its impacts.

Given the exponential uptake of generative AI that is fundamentally altering the paradigm of machine-generated content, this study begins with an assumption that even recent scholarship is already out of date. In this paper, building on a 2021 systematic review by Danzon-Chambaud, we present the results of our comprehensive and up-to-date systematized review of automated journalism scholarship, which found a six-fold increase in the number of studies in just 4 years, identifying 185 peer-reviewed studies published between 2012 and 2024, alongside an explosion of terms used to describe AI- and algorithmically generated news, illustrating significant conceptual fragmentation in this rapidly evolving area.

2. Literature Review

In this section, we briefly review the development of automated journalism. We highlight the implications of the rapid rise of AI in general, and generative AI in particular, in both reshaping journalistic practice and norms and complicating the study of automated journalism.

2.1. Early Automation

Digital technologies have fundamentally changed how we interact with information, and with each other, and have ultimately changed the paradigm of journalism which previously existed in a news landscape marked by scarcity, with a limited number of outlets, and a limited amount of content that could be circulated at any given time (Carlson, 2015; Jarvis, 2014). As a result of the digitization of our information ecosystem, traditional media outlets no longer hold either of two longstanding monopolies: the production and the distribution of information (Jarvis, 2014). Collectively, the implications of digitization—lower costs, increased access and speed, networked communication, and lost monopolies—have unraveled the journalism industry over the past two decades (Alexander et al., 2016; Benson, 2018; Kovach & Rosenstiel, 2021; Wilkinson & Winseck, 2019). In particular, Hamilton and Turner (2009) note that it is ubiquitous computation that has “transformed the landscape of journalism.” Journalists have always used tools—notepads, voice recorders, telephones, typewriters, fax machines, computers, email, databases, and content management systems—to stretch and expand their capacity to collect, document, present, and spread information, making the work faster and easier.

Scholars have previously examined the impacts of the printing press, telegraph, radio, TV, satellites and the internet on the institution of journalism. Now researchers are turning their attention to digital technologies, and the sphere of computational journalism, an umbrella term under which a number of different “journalisms” fall, including algorithmic journalism, data journalism, and automated journalism. Diakopoulos (2019) first defines computing as “the systematic study of algorithmic processes that describe and transform information,” where an algorithm is “a series of steps that is undertaken in order to solve a particular problem or to accomplish a defined outcome.” Algorithms are not inherently digital; the inverted news pyramid, for example, is an algorithm. Initially, Hamilton and Turner (2009, p. 2) described computational journalism as “the combination of algorithms, data, and knowledge from the social sciences to supplement the accountability function of journalism.” A decade later, Diakopoulos (2019, p. 27) offered a slightly refined version:

“information and knowledge production with, by, and about algorithms that embraces journalistic values.”

2.2. AI and Automated Journalism

While computational tools have previously included desktop computers, laptops, cell phones, smartphones, networked devices, and a range of software and applications that have made journalism faster and easier, artificial intelligence presents an entirely new paradigm of computational journalism. AI is a subfield of computer science which itself boasts a number of subfields, including machine learning and natural language processing (NLP) (Broussard et al., 2019). While AI is leveraged in vastly different ways across various disciplines, leading to a range of descriptions and definitions, it broadly refers to “processes in which technology simulates human intelligence and that allows for computers and machines to behave in a similar way to people” (Peña-Fernández et al., 2023, p. 2). Within the context of journalism, AI has been deployed across all phases of the news cycle, including (1) newsgathering, where AI tools help reporters sort through huge amounts data, do event monitoring, act as automated assistants, and transcribe and translate source materials; (2) production, where in addition to automated content, AI is used for summarization, to generate headlines and social media, for data visualization and article translation; and, (3) distribution, where AI optimizes and personalizes web content, interacts with news audiences through chatbots, and adjusts subscription rates and access (Aissani et al., 2023; Broussard et al., 2019; Dodds et al., 2025; Ioscote et al., 2024; Verma, 2024).

AI is becoming an increasingly key component of automated journalism, which has most commonly been understood as the use of algorithms to convert raw data into text that mimics the writing styles of human journalists (Carlson, 2015; Danzon-Chambaud, 2021; Kotenidis & Veglis, 2021). Initially, automated journalism was similar to a high-tech version of Mad Libs, where structured data—data that have been sorted and categorized, generally in a tabular format—are automatically inserted into pre-written templates (Diakopoulos, 2019) and are most often used for number-heavy content like weather forecasts, financial reports, sports statistics, and election results. This process can be entirely automated, but is more commonly semi-automated, where at minimum editors review and sign off on machine-generated drafts, or where reporters use it as a base to build on, adding descriptive narratives and additional reporting. While a number of news organizations have deployed automated journalism tools internally, such as the LA Times’ Quakebot, the Washington Post’s Heliograf and Reuters’ Lynx Insight, these technologies are more often purchased from external service providers and customized (Schapals, 2020). A few common tools used to produce automated journalism include Automated Insights’ Wordsmith, RADAR News, Retresco, Aria NLG and, previously, Narrative Science’s Quill.

As these technologies evolve and capitalize on generative AI’s ability to process unstructured data—for example narrative descriptions, images and video—and generate unique synthetic content, the breadth and complexity of the journalistic content that can be generated will also evolve, representing a growing area of research (Arguedas & Simon, 2023; Cools & Diakopoulos, 2024; Guzman & Lewis, 2024; Shi & Sun, 2024).

2.3. Generative AI

Generative AI, sometimes abbreviated to GAI or GenAI, is an umbrella term used to describe complex AI systems that use vast amounts of data to create new content, including text, images, audio, and video, known as synthetic content (Arguedas & Simon, 2023; Cools & Diakopoulos, 2024). The exponential uptake of generative AI tools was ushered in by OpenAI in November 2022, when the American tech firm unveiled ChatGPT, an

AI-driven chatbot that can process human prompts and uses vast datasets to generate text, images and computer code in response (Shi & Sun, 2024). Since then, generative AI has become ubiquitous, with major technology firms deploying their own versions, from Google's Gemini to Microsoft's Copilot. Social media platforms have embedded generative AI tools, such as Meta AI, TikTok's Symphony and X's Grok, while ChatGPT alternatives like Anthropic's Claude or Perplexity are growing in number. Unlike previous AI-enabled tools, which helped journalists work more efficiently, generative AI introduces a fundamentally different capability: "In contrast to narrow AI applications designated to carry out a specific task, generative AI tools mark a step-change that seems particularly disruptive to media enterprises because they can create content—text, images, audio, video, code, and more—that seem uncannily humanlike" (Guzman & Lewis, 2024, p. 348).

While techno-optimists argue that these powerful tools further enhance journalists' capabilities, personalize content in ways that can engage fragmented news audiences, make news and information more global and accessible, and will ultimately serve to revitalize journalism (Munoriyarwa & de-Lima-Santos, 2025), many argue that the arrival of generative AI presents an existential threat unlike any to date. From AI slop and pink slime journalism—prolific, low-quality AI-generated content masked as legitimate news sources (Darr, 2023; Goudarzi, 2024)—to deep fakes and synthetic audio, there is widespread and growing concern that the outputs of generative AI will have profound effects not just for journalism but for peoples' understanding of truth and facts, their trust in social institutions, and ultimately the future of democracy (Arguedas & Simon, 2023; Cools & Diakopoulos, 2024; Munoriyarwa & de-Lima-Santos, 2025).

2.4. Systematically Reviewing Automated Journalism Scholarship

In 2021, Danzon-Chambaud published a review of automated journalism scholarship, which identified 33 empirical studies published in English between 2012 and 2020. However, with the pace of AI development since this review was completed, alongside the uptake of generative AI, the state of automated journalism research has shifted considerably in just a few years, leaving a gap in our collective understanding of the amount and type of studies being conducted, as well as the theoretical and methodological approaches and critical issues that currently underpin this scholarship.

Exacerbating this gap is a deluge of terms at play. Danzon-Chambaud (2021) based his review on five key terms, which at the time were the most prevalent: automated journalism, algorithmic journalism, robot journalism, computational journalism, and machine-written journalism. Since then, there has been a move away from more general (computation, AI or algorithmic journalism) or contested terms (robot or robo-journalism), as well as a growing number of increasingly specific terms being used to distinguish the ever-increasing variations in this type of content, such as automated news generation (Oh et al., 2020; Wagner, 2022), auto-written news stories (Y. Wu, 2020), algorithmically generated news (Jang et al., 2022), and AI-generated news (S. Kim & Kim, 2020). Even when Danzon-Chambaud's (2021) review was published, the definition of automated journalism blurred the lines between basic automation and generative AI, and with the exponential uptake of generative AI, the distinction of what constitutes automated journalism has become even less clear. When we began this study, we had already identified at least 25 alternatives to the term automated journalism (see Table 1). This overabundance contributes to a lack of conceptual clarity that makes it challenging for scholars to have an accurate overview of the scope of existing research or direct future lines of inquiry and calls for deeper exploration into the current literature.

Table 1. Initial search terms.

Term	Source	Test Search	Refined Search String
automated journalism			automat* AND journalis*
AI authorship	(Henestrosa et al., 2023)	2	
AI journalism	(Moravec et al., 2020)	37	
AI news	(Lee et al., 2020)	27	AI AND news
AI-generated news	(S. Kim & Kim, 2020)	2	
artificial journalism	(Túñez-López et al., 2019)	57	artificial AND journalis*
algorithm authorship	(Tandoc et al., 2020)	2	
algorithm-generated news	(Y. Kim & Lee, 2021)	429	algorithm* AND news
algorithm-written news	(Jang et al., 2021)	154	algorithm* AND news
algorithmic journalism	(Dörr, 2016)	25	algorithm* AND journalis*
algorithmically generated news	(Jang et al., 2022)	22	algorithm* AND news
auto-written news stories	(Y. Wu, 2020)	1	
automated content production	(Kotenidis & Veglis, 2021)	3	
automated journalistic writing	(Y. Wu, 2020)	2	
automated news	(Haim & Graefe, 2017)	65	automat* AND news
automated news content generation	(Y. Kim & Lee, 2021)	13	automat* AND news
automatic news article generation	(Klimashevskaja et al., 2021)	0	
automatically produced content	(Galily, 2018)	1	
news automation	(Danzon-Chambaud, 2023)	43	automat* AND news
computational journalism	(Cools et al., 2022)	52	computational journalis*
computer-generated news	(Graefe et al., 2016)	2	
machine authorship	(Waddell, 2018)	0	
machine-written journalism	(Danzon-Chambaud, 2021)	1	machine AND journalis*
machine-written news	(van Dalen, 2012)	1	
robot journalism	(Clerwall, 2014)	28	robot journalis*
robo-journalism	(Blankespoor, 2018)	2	

As such, this study picks up where Danzon-Chambaud's (2021) review left off. In response to the rapid development and the fundamental changes presented by generative AI, we expand on this review by developing more nuanced search parameters in order to capture and assess a fuller range of existing automated journalism scholarship. In addition to empirical studies, we also include conceptual, applied, and analytical research, as well as empirical studies, to build a more fulsome picture of automated journalism scholarship. Finally, we add to our analysis a summary of prevalent theories, concepts and frameworks.

As such, the research questions guiding this systematized review are as follows:

- RQ1 What patterns arise from an updated and expanded systematic retrieval of peer-reviewed automated journalism scholarship?
- RQ2 How does an expanded and more nuanced selection of key search terms impact the results?
- RQ3 What theoretical and conceptual frameworks are most prominent when a systematic review is expanded beyond empirical studies?

3. Methods

This study comprises a systematized review of automated journalism scholarship, as described by Grant and Booth (2009) in their typology of reviews. This type of research synthesis includes elements of a systematic review, best suited to assess the quantity and scope of available research, and to characterize key features (Grant & Booth, 2009). Unlike a systematic review, a systematized review does not require a formal quality assessment of included studies and is instead meant to summarize research, identify existing gaps, and refine future lines of inquiry. The steps we follow include: identifying appropriate databases, designing and testing a search strategy, running the searches and exporting the

results, a title and abstract screening to eliminate irrelevant studies, a full text review to include or exclude the remaining studies, and finally, analysis of the resulting corpus.

While this systematized review falls under disciplines outside of the health sciences and does not include a quality assessment of included studies, it has been aligned as closely as possible to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 Statement (Page et al., 2021). The completed 27-item PRISMA 2020 Checklist can be found in Appendix B.

3.1. Search Strategy

Prior to conducting the search, and at regular intervals throughout data collection, a research librarian was consulted to design and operationalize a formal search strategy, including determining the appropriate search criteria, parameters and sources, designating inclusion and exclusion criteria, testing and validating search strings, identifying validation articles, and updating the initial search.

3.1.1. Inclusion Criteria

A search strategy was designed to identify studies with the following characteristics: (1) peer-reviewed studies, (2) published in English, and (3) about machine-generated textual news content published online. While Danzon-Chambaud's (2021) corpus was limited to empirical studies, this review was extended to include articles that were also theoretical, applied, or analytical, to identify gaps in research, critical issues that have not yet been studied empirically, technical studies, and other research synthesis.

This review also includes studies that are fully about automated journalism, as well as those that are partially about automated journalism (not solely about automated journalism, but includes a significant portion on the topic), or tangentially about it (directly relevant to the subject of automated journalism, even if the paper itself is not explicitly about automated journalism). Finally, we chose to include studies that were clearly touching on the topic of automated journalism but were poorly written, poorly translated, or where the method or descriptions were challenging to understand; these were tagged as unsure.

3.1.2. Databases

Four social science databases were chosen for this study, on the advice of the consulting research librarian: (1) EBSCO's Communication Source, the most prevalent database for communication and media scholarship, indexes 79 journalism-related journals and 278 media journals; (2) EBSCO's Academic Search Complete provides a broad scholarship overview, with indexing of potentially related disciplines outside of the field of communications and media, such as political science, sociology, philosophy and psychology; (3) all six indexes of the Web of Science—in addition to both social sciences and arts and humanities indexes, Web of Science also includes a science citation index and conference proceedings, which could return related results from the fields of computer science and engineering, as well as an index of emerging journals, which may capture studies from more novel or exploratory fields like technology; and, (4) SCOPUS—a well-established counterpart to Web of Science, providing some overlap.

3.1.3. Search Terms

This review was also expanded to include a wide variety of terms increasingly being used to describe machine-generated news content. We found 25 variations of the term automated journalism identified within a preliminary search, which we used to build and refine our search strategy. Each term was tested individually in the Communication Source database to initially assess the number and relevance of the results (see Table 1).

Search terms with limited results were eliminated, and similar terms were then aggregated, with Boolean operators and wildcards added, to create a more refined set of 10 search strings (See Table 1). These were tested again for relevance and breadth, and refined to the following final list of search strings:

1. automat* nX (journalis* OR news).
2. algorithm nX (journalis* OR news).
3. AI AND (journalis* OR news).
4. “artificial intelligence” AND (journalis* OR news).
5. computational journalis*.
6. robot journalis*.
7. machine journalis*.

Each of these search strings was adjusted for syntax and tested individually in each of the four databases, adjusting proximity indicators—a search operator which specifies how far apart two search terms can be—until returns no longer included relevant results. Once each search string was finalized for each database, all seven were combined into a single search string. For example, the final search string for Communication Source is as follows:

(automat* n5 (journalis* OR news)) OR (algorithm n6 (journalis* OR news)) OR (AI AND (journalis* OR news)) OR (“artificial intelligence” AND (journalis* OR news)) OR computational journalis* OR robot journalis* OR machine journalis*)

3.2. Data Collection

An initial search limited to English-language, peer-reviewed studies was conducted on 1 May 2024, and updated on 27 January 2025, to collect complete data for 2024. Search results were exported from each of databases in .ris format and imported into Covidence, an online systematic review tool. A total of 3211 articles were imported, and 878 duplicates were automatically removed, leaving 2270 articles to be screened. The titles and abstracts were manually screened to eliminate irrelevant studies, including those outside of the discipline, non-English studies, and non-peer reviewed articles, such as book reviews and commentary. Through screening, 1807 studies were deemed irrelevant and an additional 63 duplicates were manually identified.

Two authors then independently reviewed all 463 articles remaining for full-text review, choosing which to exclude based on the following criteria: (1) not automated journalism; (2) not textual automated journalism; (3) not peer-reviewed; (4) not in English; or (5) general overview. Studies were coded as general overviews when they were not explicitly about automated journalism, but rather about the broader impact of digital technologies on the journalism domain, wherein automated journalism may have been listed as an example but not further discussed. One retracted article was also excluded. Once the remaining studies were reviewed independently, the authors met to discuss any disagreements, and a final decision was made by consensus (Syed & Nelson, 2015; Zade et al., 2018), resulting in 278 studies excluded. The final corpus includes a total of 185 studies, which are listed in Appendix A. The PRISMA flow chart generated by Covidence is shown in Figure 1.

Of note is that all 33 Danzon-Chambaud’s studies included in his 2021 review were accounted for in our results. However, we chose to exclude two of these studies, which were about Chatbots and Twitter bots, as they did not meet our current definition of automated journalism.

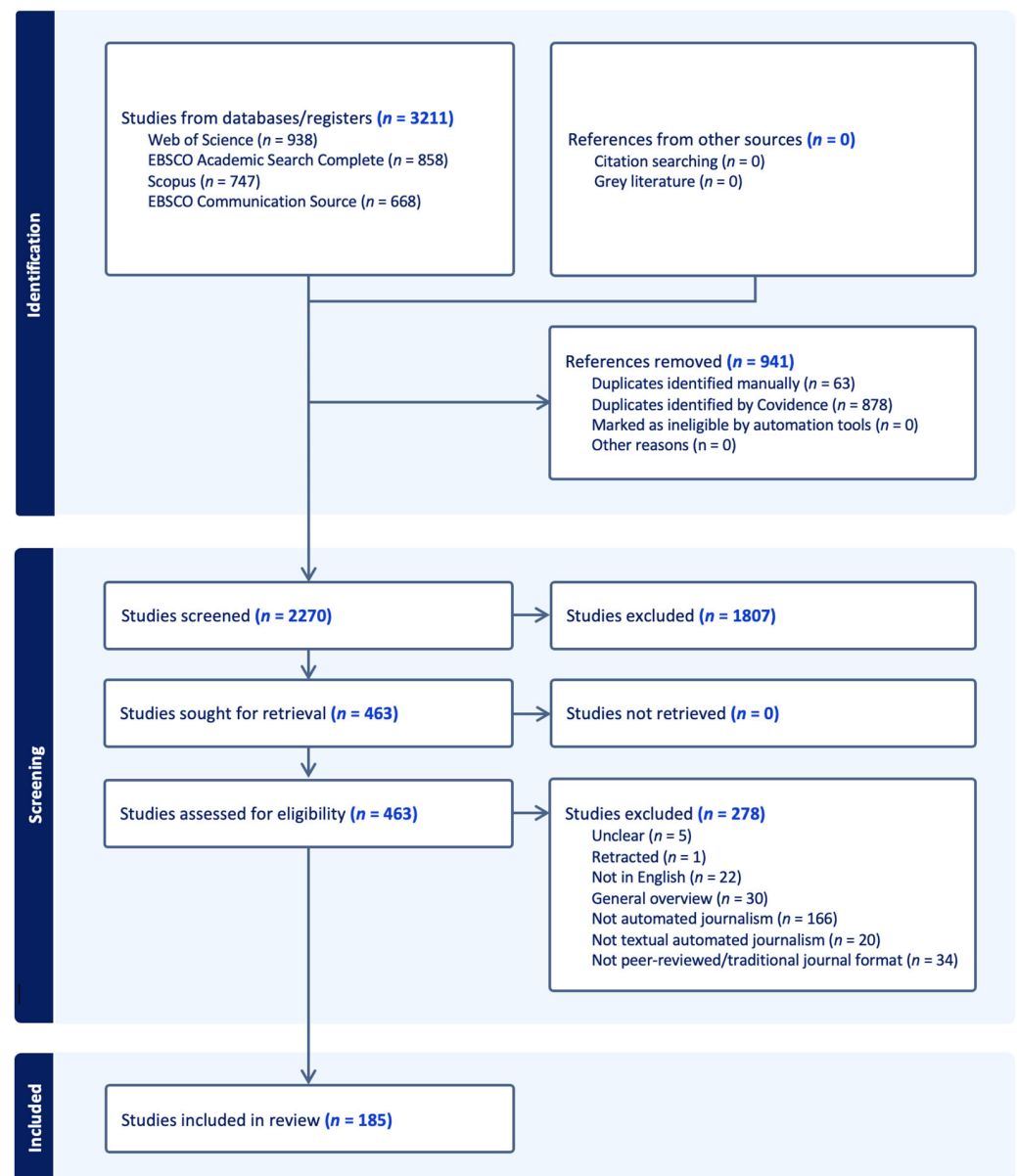


Figure 1. PRISMA flow chart.

3.3. Data Analysis

The final corpus was exported from Covidence in tabular format and then imported into the online database tool AirTable for analysis. A codebook was developed outlining each element of analysis including study type; keywords; authors and countries of origin; journals and disciplinary areas; regions of study; news organizations; theoretical frameworks; research methods and tools; and automated journalism terms and tools. The analysis also included more qualitative notes on study findings, critical issues, and areas of future research.

Both authors initially independently analyzed and coded the first 10% ($n = 15$) of the studies included from the dataset collected on 1 May 2024 ($n = 143$). The authors then met to compare results, and any differences were again reconciled by consensus (Syed & Nelson, 2015; Zade et al., 2018). The remaining studies were then divided between the authors to be coded individually.

4. Results

In this section, we present a range of top-level results emerging from our data collection process. Responding to our research questions, we first discuss the broad patterns that arise from our updated and expanded systematic retrieval of peer-reviewed automated journalism scholarship (RQ1). Next, we examine the impacts of an expanded selection of key terms (RQ2), and finally discussing prominent theories, concepts and frameworks (RQ3).

4.1. Patterns Arising from an Updated and Expanded Review (RQ1)

We first consider the types of studies, publication dates, top journals, keywords, prevalent authors and their institutional regions, regions of study, and prevalent research methods.

4.1.1. Study Tags and Types

We included a total of 185 studies in our final corpus, of which 70% ($n = 130$) were fully about automated journalism, 17% ($n = 30$) partially about automated journalism, and the remaining 13% ($n = 25$) tangentially about automated journalism. Of these, 61% are empirical studies ($n = 112$), 11% are analytical ($n = 20$), 9% are conceptual ($n = 16$), and 5% are applied ($n = 10$). The remaining 15% of the studies were mixed ($n = 27$) and include more than one study type. See Figure 2.

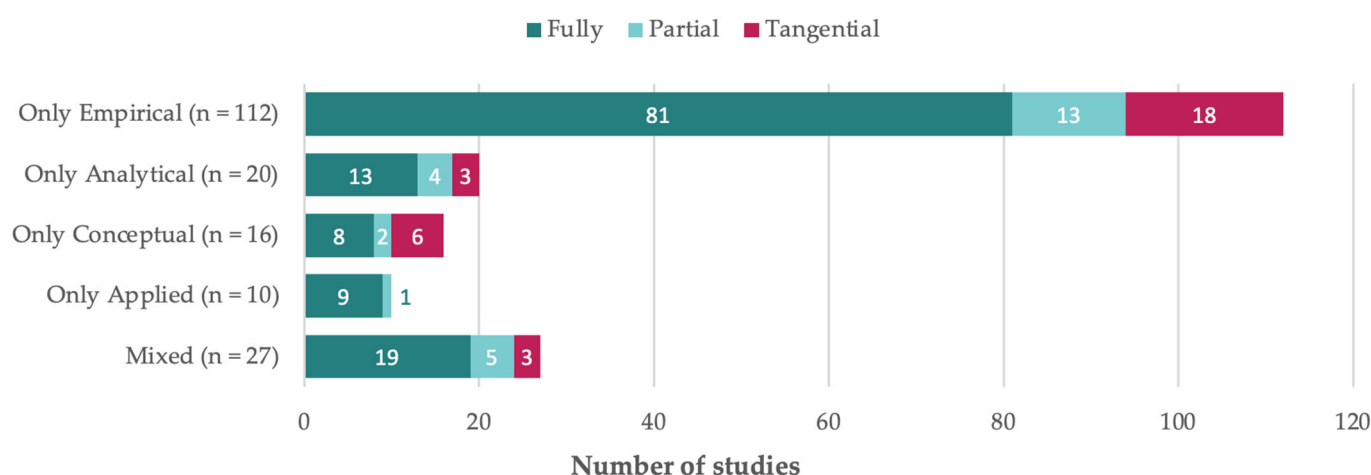


Figure 2. Study tags across study types.

4.1.2. Publication Dates and Journals

Danzon-Chambaud's (2021) study found 33 empirical studies conducted between 2012 and 2020. While we cannot compare directly, as we have included applied, analytical, and conceptual studies alongside empirical research, as well as studies that are also partially or tangentially about automated journalism, Figure 3 still shows the growth trends over time, alongside a sheer explosion of studies in 2024.

The 185 studies in our corpus were published in 73 individual academic journals, as well as in conference proceedings and lecture notes. While the top five journals in our study fell under the discipline of communications within social sciences, we in fact included studies from 46 subjects within 14 disciplines, compared to Danzon-Chambaud's (2021) study with 13 academic journals from 19 subjects within 7 disciplines. Moreover, 70% ($n = 23$) of the 33 studies were published in journalism studies journals, in contrast to only 40% ($n = 76$) of our 185 studies. See Table 2.

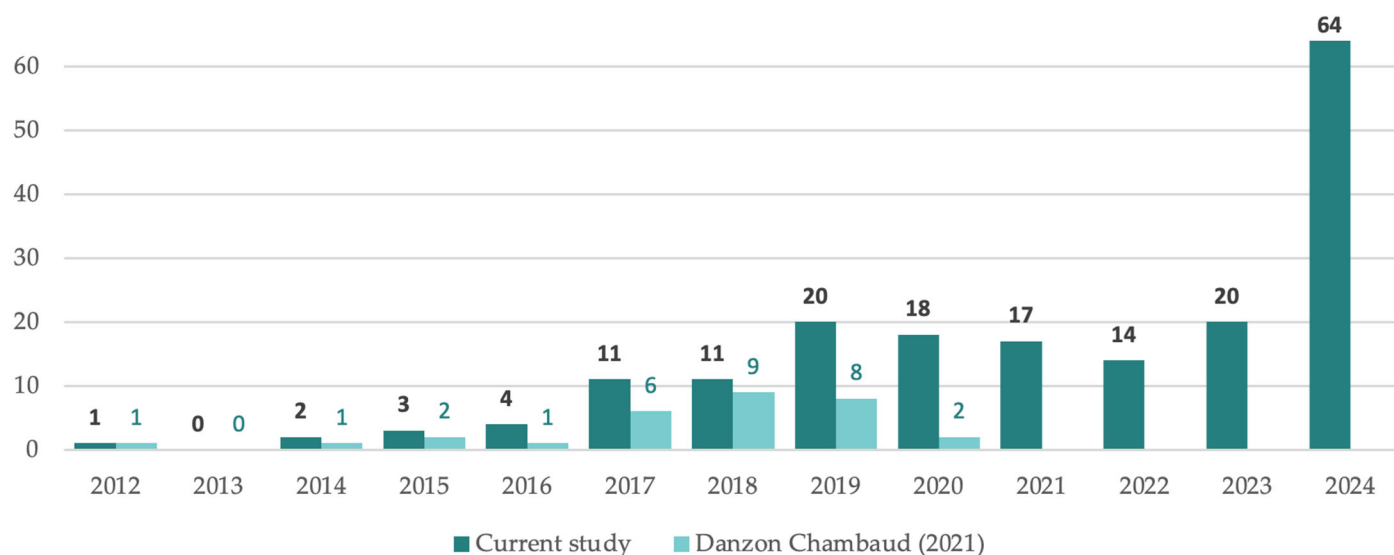


Figure 3. Number of studies by year of publication. Note: While we referred to the journal publication dates cited for each study within our corpus, the Danzon-Chambaud (2021) study used the first published date, meaning there is some variance between the records. In order to draw comparisons, we have adjusted the dates of the Danzon-Chambaud studies in this table to reflect the publication dates indicated within citations. We have also not included the three studies that we excluded for our study (one in 2016 and two in 2019), thus showing a total of 30 records, rather than the 33 in the study.

Table 2. Top journals.

Journal	Count	Publishing Country	Discipline (According to scimagojr.com)
Digital Journalism	25	United Kingdom	Communication
Conference Proceedings	18		
Journalism Practice	16	United Kingdom	Communication
Journalism	14	United Kingdom	Arts and Humanities (miscellaneous)
Journalism and Media	8	Switzerland	Arts and Humanities (miscellaneous)
Journalism Studies	7	United Kingdom	Linguistics and Language Communication

4.1.3. Keywords

The author-listed keywords from each study were captured, with 531 unique keywords emerging from a total of 1030. Table 3 includes the top raw keywords and a list of aggregated groupings of keywords. For example, the automated journalism grouping includes related terms, such as automated journalistic writing, automated news, automated news generation, automated news writing, automated text generation, Chinese automated journalism, machine generated news, and machine-written news.

Table 3. Top keywords and keyword groupings.

Raw Keywords	Count of Studies	Keyword Grouping	Count of Studies
automated journalism	66	automated journalism	99
artificial intelligence	57	artificial intelligence	65
journalism	43	journalism	48

Table 3. *Cont.*

Raw Keywords	Count of Studies	Keyword Grouping	Count of Studies
robot journalism	33	robot journalism	41
computational journalism	20	algorithmic journalism	24
algorithmic journalism	19	credibility	22
automation	18	ethics	22
automated news	13	computational journalism	21
credibility	12	algorithm(s)	20
news	12	automation	18
algorithms	11	natural language processing	17
natural language generation	11	media	16
news production	10	technology	16

4.1.4. Authors

The corpus includes 351 individual authors (compared to 70 in [Danzon-Chambaud's \(2021\)](#) study), representing institutions in 43 countries (15 in [Danzon-Chambaud \(2021\)](#)), with more than half coming from five countries (see [Table 4](#)): the United States ($n = 76$, 21.7%), Spain ($n = 36$, 10.3%), South Korea ($n = 27$, 7.7%), Germany ($n = 25$, 7.1%), and China ($n = 18$, 5.1%).

Table 4. Top Countries of Authors.

Country	Count of Authors ($n = 351$)	
United States	76	21.7%
Spain	36	10.3%
South Korea	27	7.7%
Germany	25	7.1%
China	18	5.1%
United Kingdom	17	4.8%
Finland	15	4.3%
India	12	3.4%
Netherlands	11	3.1%
Norway	9	2.6%

Of the 351 individual authors, 14 authored at least four studies within the corpus. See [Table 5](#).

Table 5. Top Authors.

Author	Country of Institutional Affiliation	Count of Studies ($n = 185$)
Konstantin Nicholas Dörr	Switzerland	5
Chenyan Jia	United States	5
Edson C. Tandoc Jr.	Singapore	5
Shangyuan Wu	Singapore	5
Samuel Danzon-Chambaud	Ireland	4
Laurence Dierickx	Belgium	4
Daewon Kim	South Korea	4
Soojin Kim	United States	4
Leo Leppanen	Finland	4
Carl-Gustav Linden	Finland	4
Václav Moravec	Czech Republic	4
Colin Porlezza	Switzerland	4
Sina Thäsler-Kordonouri	Germany	4
Neil Thurman	Germany	4

4.1.5. Regions of Study

Within our 185 studies, research was conducted in 41 countries, as well as across eight distinct regions (Europe, Africa, Latin America, Middle East, The Caribbean, the Arab World, South Asia, Asia Pacific, and Catalonia). There were 24 countries within which individual studies were conducted (Figure 4), and another 17 countries represented comparative studies. The top 5 countries overall were the United States ($n = 51$), the United Kingdom ($n = 22$), Spain ($n = 20$), Germany ($n = 19$), and South Korea ($n = 12$).

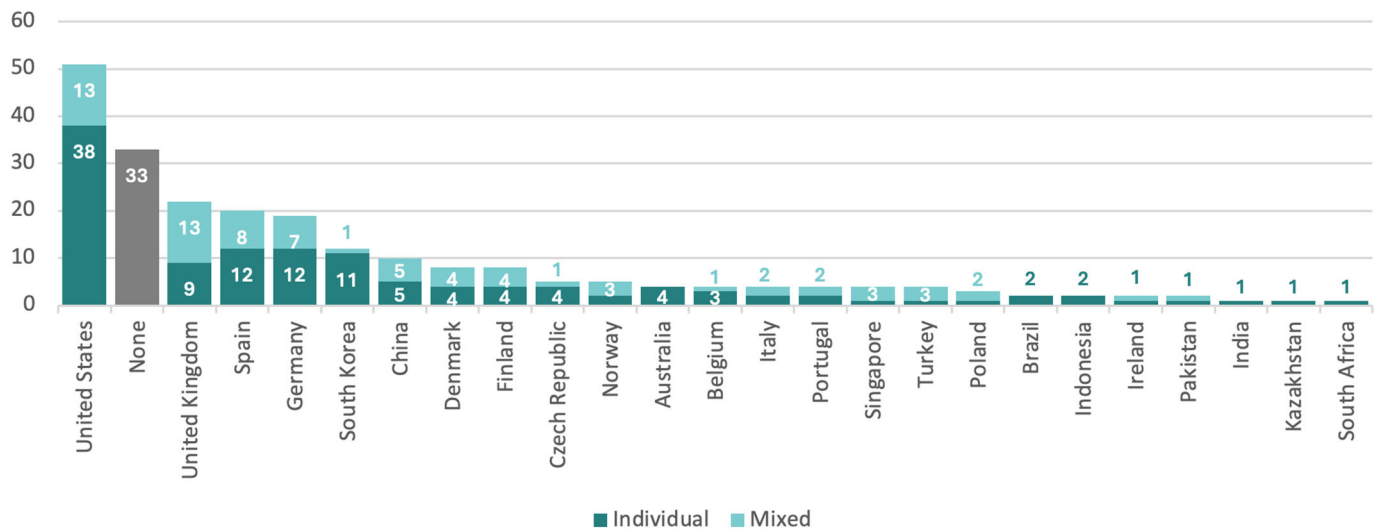


Figure 4. Regions of individual study.

In addition to studies conducted in individual countries, another 17 countries were part of comparative studies, including Austria ($n = 4$), Bulgaria ($n = 1$), Canada ($n = 2$), Estonia ($n = 2$), France ($n = 6$), Greece ($n = 1$), Hungary ($n = 1$), Iceland ($n = 1$), Japan ($n = 1$), Luxembourg ($n = 1$), Netherlands ($n = 2$), Romania ($n = 1$), Slovenia ($n = 1$), Sweden ($n = 3$), Switzerland ($n = 4$), Taiwan ($n = 1$), and Ukraine ($n = 1$).

4.1.6. Research Methods and Tools

To evaluate the research methods within our corpus, we captured methods as described by the authors, then grouped them into types. With the 185 studies, we found 444 total mentions of 157 unique methods, which we grouped according to common research types, shown in Table 6. Recognizing that individual studies mention the use of multiple methods, and that the count totals are not proportional representations across the corpus, we found 35% of the studies in our corpus noted qualitative methods ($n = 66$), including 52 studies that conducted interviews, 18 case studies, 7 using ethnographic approaches and 3 focus groups. One quarter of the studies used experimental methods ($n = 45$), with at least 17 conducting between-subject tests, which were generally one of two themes: (1) testing whether audiences could tell the difference between automated- and human-generated content; and (2) accessing audience perceptions of automated- versus human-generated content. About 4% of the studies are applied research, which were most often the description of a computational technique to produce automated news content. A quarter of the studies note using surveys ($n = 47$), and almost 10% describe variations in systematic reviews ($n = 17$). When it comes to analyses, 30% of the studies applied content analysis techniques ($n = 54$), while 16% performed statistical analysis. Finally, 12% of the studies in our corpus described other research designs ($n = 22$), such as theoretical or philosophical approaches like grounded theory or the dialectic method, while 10% describe

their sampling approaches ($n = 20$), most often snowball sampling or purposive sampling. See Table 6.

Table 6. Research methods.

Types	Individual Studies ($n = 185$)	
Qualitative methods	66	35.7%
Interviews	52	28.1%
Case study	18	9.7%
Ethnography	7	3.8%
Focus Group	3	1.6%
Content analysis	54	29.2%
Thematic analysis	21	11.4%
Content analysis	14	7.6%
Qualitative data analysis	8	4.3%
Discourse analysis	5	2.7%
Legal analysis	4	2.2%
Survey	47	25.4%
Online survey	28	15.1%
Manipulation check	6	3.2%
Experimental design	45	24.3%
Online experiment	30	16.2%
Between-subject	17	9.2%
Statistical analysis	30	16.2%
Other research approaches	22	11.9%
Sampling	20	10.8%
Snowball sampling	8	4.3%
Purposive sampling	6	3.2%
Systematic review	17	9.2%
Applied	8	4.3%

Note: Each count represents the number of studies that reported applying a particular research method, or a subcategory of a method. Individual studies mention the use of multiple methods, meaning the table reflects raw counts of studies.

In addition to documenting research methods, we also noted 139 unique tools mentioned 233 times within the studies (Table 7). One quarter ($n = 34$) were applications, in particular research software such as NVivo, Zoom, SPSS, and MaxQDA. Another 20% ($n = 27$) were databases, including research databases like Scopus, Lexis Nexis, and Web of Science, as well as public databases like Google Scholar, Compustat, or MediaCloud. Computational techniques like coding languages comprised 10% ($n = 15$) and another 10% were survey tools like Qualtrics and YouGov.

Table 7. Top Research Tool Categories.

Category	Example	Number of Tools ($n = 139$)	
Software & apps	NVivo, Zoom, SPSS, MaxQDA, Skype, Excel	34	24.5%
Database	Scopus, Lexis Nexis, Web of Science, Google Scholar, MediaCloud	27	19.4%
Computation & coding	Python, APIs, HTML, JavaScript	15	10.8%
Survey Tools	Prolific, Qualtrics, SoSci Planet, YouGov	10	7.2%

Table 7. Cont.

Category	Example	Number of Tools (<i>n</i> = 139)	
Measures	Computer Attitude Scale (CAS), Trust in News Media scale	9	6.5%
Language models	Chat GPT, Chat Open AI, GPT, Paddle NLP, Google GeminiAI	7	5.0%
Platforms	OSF, Twitter, LinkedIn, WeChat, Weibo, Facebook, Slack	7	5.0%
Dataset	AP Index, ActivityNet Captions, RealNews Dataset	6	4.3%
AI tools	StatsPerform, HeyWire AI	4	2.9%
Search Engines	Google, ResearchGate	4	2.9%
Recruitment	Prolific, Amazon Mechanical Turk, Research Match	3	2.1%
Other	Including organizations, projects, repositories, websites	13	9.4%

By raw count within the 185 studies, the five most prevalent tools are Amazon’s crowdsourcing tool Mechanical Turk (*n* = 13), the academic database Scopus (*n* = 10), the AI tool Prolific (*n* = 9), the qualitative content analysis software NVivo (*n* = 8) and the survey tool Qualtrics (*n* = 8).

4.2. Expanded Selection of Search Terms (RQ2)

As previously described, the wide range of terms in use as alternatives to automated journalism necessitated a rigorous and nuanced search strategy meant to capture any number of combinations of terms. In addition to author-provided keywords (shown in Table 3), we compiled a list of terms used interchangeably with “automated journalism” within our corpus. For example, some authors may have used the term “automated journalism” in their title and keywords but additionally used related terms like “automated news” and “automated content” throughout their text. To our surprise, we documented 157 overlapping but distinct terms mentioned 572 times across our 185 studies. We also noted an additional 11 terms used in studies that were ultimately excluded from our corpus. The top terms aligned almost exactly with [Danzon-Chambaud’s \(2021\)](#) search terms: 64% of the studies mentioned the term automated journalism (*n* = 119), 35% mentioned robot journalism (*n* = 65), 26% mentioned algorithmic journalism (*n* = 48), and 22% mentioned computational journalism (*n* = 40). Because there were often slight but distinct variations—for example, algorithm journalism as well as algorithmic journalism, or automated journalism, automatic journalism and automated journalistic outputs—we also grouped such similar terms together, which increased the prevalence of automated journalism variations to 68% (*n* = 185), and more than doubled variations of automated news to 32% (*n* = 59). With this grouping, we also start to see the prevalence of AI-generated (content, text, journalism, articles) and AI journalism (both *n* = 12). See Table 8.

Table 8. Top raw terms and groupings of automated journalism terms.

Raw Term	Count of Studies (<i>n</i> = 185)		Term Grouping	Count of Studies (<i>n</i> = 185)	
automated journalism	119	64.3%	automated journalism	126	68.1%
robot journalism	65	35.1%	robo(tic) journalism	82	44.3%
algorithmic journalism	48	25.9%	algorithm(ic) journalism	61	33.0%
computational journalism	40	21.6%	automated news	59	31.9%
automated news	23	12.4%	computational journalism	40	21.6%
algorithm journalism	12	6.5%	machine-written (. . .)	18	9.7%
machine-written news	11	5.9%	automated content	17	9.2%
news automation	10	5.4%	AI-generated (. . .)	12	6.5%
robo-journalism	7	3.8%	AI journalism	12	6.5%
automated content	6	3.2%	automated text (. . .)	11	5.9%

Because so many general components of the terms were used interchangeably, we also broke down the elements of each of the 157 terms (for example automation + journalism) into 52 components. The most prevalent components are shown in Table 9, along with examples, which provide a clearer picture of how these elements are used interchangeably to generate dozens of alternative terms.

Table 9. Breakdown of term elements.

Term Element	Examples	Count (<i>n</i> = 157)
automated, automation, automatic(ally)	automatic journalism automated production of news automatically generated articles automated computer-written articles	63
news	machine-written news news automation automated news production computer-generated news content	54
journalism, journalist, journalistic	machine-written journalism automated journalistic writing AI-generated journalistic content machine-produced journalism	45
generation, generated, generative	AI-generated articles automated journalistic text generation automatically generated news stories autogenerated news	36
AI, artificial intelligence	AI-driven content generation AI-written media texts artificial intelligence journalism AI-authored articles	34
writing, written	AI-written news (articles) automated computer-written articles algorithm-written news (stories) machine-written journalism	34
content	automated content production machine-generated content automated production of journalistic content computer-generated news content	26
algorithm(s), algorithmic, algorithmically	algorithmic news production algorithm-generated stories automated news-making algorithms algorithmically assembled news	24
produced, production	automated content production machine-produced journalism algorithmically produced news AI-based news production	21
text(s), textual	automated text generation AI-written text(s) automatically generated textual news automated text production	21

4.3. Theories, Concepts and Frameworks (RQ3)

One area of analysis that we wanted to give more consideration to was the theoretical and conceptual approaches and frameworks underpinning the studies in our corpus. These were generally identified by topics addressed within literature review sections of the studies. However, because automated journalism is also an applied practice, this resulted in a broad and varied list that not only included theoretical and conceptual frameworks, but also

applied concepts, models, and research areas. We found a total of 242 distinct topics, of which the top three accounted for less than 3% of each study (see Table 10).

Table 10. Top terms.

Term	Count of Studies (<i>n</i> = 185)	
news credibility	12	2.80%
natural language processing (NLP)	10	2.33%
natural language generation (NLG)	10	2.33%
MAIN model theory	9	2.10%
machine heuristic (MH)	9	2.10%
human–machine communication	8	1.86%
large language models (LLM)	8	1.86%
Bourdieu Field Theory	7	1.63%
journalism ideology	7	1.63%
journalism ethics	7	1.63%

Given this broad range, we chose to group the 242 terms into three categories: (1) concepts, which includes academic terms and definitions (*n* = 103, 43%); (2) theories and frameworks, which include developed theoretical explanations or frameworks applied academically (*n* = 79, 33%); and (3) applied practices, which include concepts or models that are operationalized in applied settings (*n* = 60, 25%). Additionally, given the limited prevalence of any one term, we also chose to group similar terms and count them together. For example, under concepts, a number of terms relating to news and media credibility were grouped together, including news credibility, source credibility, message credibility, and credibility perception. Under theories, variations of hostile media theories, including hostile media bias (HMB), hostile media effect (HME), and hostile media perceptions (HMP), were counted together, while in applied practices, natural language processing and natural language generation were bundled.

Even with these groupings, with the exception of news & media credibility, which was mentioned in 14% of our corpus' 185 studies (*n* = 26), none accounted for more than 10% of the studies in our corpus, indicating a very broad range of theoretical and conceptual approaches with the study of automated journalism (see Table 11).

Table 11. Top theoretical and conceptual approaches and frameworks.

Concepts	Count of Studies (<i>n</i> = 185)	
News & media credibility	26	14.1%
Human–computer interaction/collaboration	19	10.3%
Heuristics	11	5.9%
Transparency/Disclosure	9	4.9%
Journalistic identity/Role conceptions	8	4.3%
Institutional logics	8	4.3%
Theories & Frameworks	Count of Studies (<i>n</i> = 185)	
Hostile media theories	11	5.9%
MAIN model	9	4.9%
Expectancy theories	8	4.3%
Philosophical frameworks	8	4.3%
Socio-technical theories	8	4.3%
Hostile media theories	11	5.9%

Table 11. Cont.

Applied practices	Count of Studies (<i>n</i> = 185)	
Natural language processing	20	10.8%
Journalism & media ethics	15	8.1%
Legal frameworks	14	7.6%
Language models	9	4.9%
AI and generative AI	8	4.3%

However, despite no overly dominant theories, concepts or frameworks, there were a number of clear themes that emerged when items were more generally clustered across the three categories. For example, concepts, theories and frameworks broadly related to journalism scholarship were grouped, such as news values and norms (applied), journalism ideology (concept) and the spheres of journalism theory (theories), while concepts of trust and credibility were grouped with the MAIN model theory, which is a framework for measuring credibility of online news sources. The prevalence of these themes was then estimated by adding up the number of times each term within a theme was mentioned across all studies in the corpus. For example, news credibility was mentioned in 12 studies, source credibility in 5 studies, trust in 5 studies, the MAIN model in 9 studies, and so on, with theories, concepts and frameworks related to credibility and trust mentioned on 44 occasions (see Table 12).

Table 12. Theoretical and conceptual themes.

Theme	Description	Prevalence
Journalism theories	journalism ideology, role conception, authority, boundaries, identity, judgement, innovation as well as meta-journalistic discourse	46
Credibility and trust	news, media, source, message and medium credibility; perceptions of credibility and related trust.	44
Machine language	Natural language processing (NLP) and natural language generation (NLG); language modeling, and related concepts such as named entity recognition (NER), semantic representation, and post-editing.	34
Socio-technical theories	technological drama, imaginaries, determinism, media effects, memory, reductionism, adoption and appropriation; socio-technical construction; the SCOT model; technology acceptance models; and technological innovation theories.	29
Institutional theories	new/neo, discursive and historical institutionalism; institutional logics, entrepreneurship and isomorphism; multifactorial resistances; structural inertia and anticipatory practices.	23
Human-machine interaction (HMI)	human-machine communication, human-computer interaction, human-artificial intelligence [AI] interaction, and human-computer collaboration.	22
AI and algorithms	Algorithmic aversion, judgment, transparency and literacy; adaptive, generative, general purpose and communicative AI; media synthesis; uncanny valley effect and word-of-machine effect.	22

Note: The numbers in the table above do not represent the exact number of studies, as a study that mentions news credibility may also be using the MAIN model, for example. It is instead meant as an estimate of the prevalence of particular themes, which are highlighted in the table above.

5. Discussion

Our updated and expanded systematized review of automated journalism scholarship shows an exponential increase in the number of studies, authors, and countries over the past 4 years (RQ1), alongside a surge of key terms in use (RQ2), and a wide range of theoretical and conceptual approaches with the study of automated journalism (RQ3). Drawing these findings together, we see clearly how the speed of AI-driven changes within

the journalism industry, alongside the seismic shifts introduced by generative AI, are leading to fragmentation as scholars try to keep up with the terms, technologies, and tools at play. This lack of conceptual clarity and consensus makes the study of automated journalism challenging and calls for common language across the diversity of disciplines contributing to our collective understanding of this increasingly complex practice.

5.1. Exponential Growth

The clearest finding of this review is the sheer quantity of studies and the exponential increase since [Danzon-Chambaud \(2021\)](#) identified 33 studies in his review, particularly in 2024, which accounts for one-third of the 185 studies we identified since 2012 ($n = 65$). Even when recognizing that [Danzon-Chambaud \(2021\)](#) only considered empirical studies that were fully about automated journalism, and filtering for those attributes, we still find a threefold increase ($n = 94$).

While Western regions continue to dominate scholarship even as generative AI lowers technical barriers for participation, there has been a substantial increase in the number of countries represented in terms of the institutional affiliation of the authors represented within the corpus (43 countries compared to 15 previously). Additionally, we tracked the geographic regions being studied, which were not specified in the original review, and found 46 individual countries alongside a number of studies conducted across broader regions, including the European Union, Latin America, and Sub-Saharan Africa.

In addition to the regions represented within our corpus, given the significant overlap between automated news content and other AI and algorithmic applications within journalism, it is worth mentioning that we identified research taking place in a number of understudied regions, which were ultimately excluded from our corpus because they fell under the category of ‘general overview.’ As mentioned, studies were coded as general overviews when they were not explicitly about automated journalism, but rather about the broader impact of digital technologies on the journalism domain, wherein automated journalism may have been listed as an example but not further discussed. This includes Egypt, Finland, Ghana, Jordan, Kazakhstan, Vietnam, Malta, Morocco, the Philippines, Poland, Russia, Saudi Arabia, Singapore, South Africa, and the United Arab Emirates.

Within our corpus studies we also observed a wide range of research methods and tools for analysis used, pointing to breadth in the methodological and analytical approaches used to explore automated journalism. This suggests that the subject is approached by a wide variety of scholars from across a range of disciplines, including technical studies from computer science and engineering, business management, psychology and social sciences, but also from increasingly far-reaching disciplines like materials science, mathematics, environmental science, and behavioral science.

Alongside the exponential increase in the number of studies from increasingly diverse regions and disciplines, a second important finding is the sheer number of terms being used to describe machine-generated news content. We found 157 overlapping but distinct terms within our final corpus, as well as an additional 11 terms in studies that we excluded. We show a breakdown of the most prevalent components in [Table 11](#), a valuable contribution for future research to help scholars make sense of the deluge of available terms to describe automated journalism.

This explosion of terms suggests a field that is developing so rapidly that scholars are simultaneously reaching for increasingly nuanced descriptions of automated journalism on the fly. Additionally, the influence of AI can easily be seen in the evolution of these terms. Of the 157 distinct terms, almost 20% ($n = 29$) include “artificial intelligence” or “AI” as a component (for example, AI journalism, AI news, AI-driven journalism, generative AI journalism), with 1 of them occurring in a 2018 study, 4 in 2020, 1 in 2021, 1 in 2022,

3 in 2023, and the remaining 19 in 2024. In comparison, only one study in Danzon-Chambaud's (2021) corpus used an alternative to automated journalism that included artificial intelligence (specifically, "artificial intelligence systems-aided production of news items," Díaz-Noci, 2020).

The exponential growth of AI in journalism in general, and generative AI in particular, has fundamentally expanded *what* counts as automated journalism and is thus moving the field far beyond template-based text generation as initially envisaged when the technology made its first foray in the journalism domain.

5.2. Lack of Consensus and Conceptual Clarity

One of the issues with trying to make sense of this dataset is the lack of disciplinary consensus and conceptual clarity that is normally relied upon by scholars in order to build on and contribute to collective knowledge. The breadth of terms being used to describe automated content is particularly problematic for two reasons. First, it makes searching for relevant literature challenging. While this is in part because the combination of terms used as alternative for automated journalism include generic elements (automated, AI, algorithmic, journalism, news, etc.), it is also because we built our own search parameters on the elements found within studies in Danzon-Chambaud's (2021) corpus, which already included 25 alternatives for the term automated journalism. Even within these search parameters, however, we had to negotiate with thousands of irrelevant results.

Second, it undermines conceptual clarity as various authors use different terms to talk about the same object of study or, vice versa, similar terms to talk about different objects of study, especially within this interdisciplinary space. For example, the idea of data-driven journalism within journalism studies describes a type of reporting that relies on computational methods to look for stories within vast datasets. However, within our corpus it was used to discuss the need for algorithmic transparency in the context of the adoption of automated and algorithmic systems within newsrooms (Cools & Koliska, 2024). In all, it can be said that the proliferation of terminology reflects a deeply unsettled conceptual landscape, driven by rapid innovations in the AI space and inconsistent usage across disciplines.

This collection of diverse terms, however, forced us to be significantly more nuanced in how we designed our search strategy. For example, in terms of raw numbers, when we replicate Danzon-Chambaud's (2021) search query in Communication Source, it returns 151 results versus the 916 found with our search query. Acknowledging that some results might be duplicates or irrelevant, this more nuanced search query still casts a wider net that allows for the possibility of identifying a greater number of relevant studies, including those from different geographical, political or cultural contexts that may be applying different terms for any number of reasons.

One surprising finding that underscores this missing conceptual clarity is a general lack of prevalent conceptual or theoretical frameworks within our corpus. While there were themes around trust and credibility, machine language and socio-technical theories, no single concept, theory or framework—with the exception of news and media credibility (6%, $n = 26$)—was present in more than 5% of the 185 studies, even when grouped together with similar terms. One explanation may be the interdisciplinary nature of this field comprising several disciplines, each subscribing to their respective bodies of literature, and theoretical and methodological traditions. While the top five journals in our study were indeed journalism studies publications (labeled as the subject of communications within social sciences on SCImago Journal Rankings), we in fact found studies about automated journalism from 46 subjects within 14 disciplines. To that end, the same problem of breadth and range repeated itself across the corpus' keywords and research methods, with dozens

of list items that can be loosely grouped together thematically but show no distinct trends or gaps. For scholars, this lack of disciplinary consensus and conceptual clarity means they are left without clear collective signposts to guide them as they develop concepts, test theories, refine research methods, analyze data, and draw conclusions about automated journalism. This in turn forces them to create their own signposts as they carve their own paths for lack of any to follow: creating their own terms, applying their own preferred theories, citing their own narrow corpus of literature, and ultimately further exacerbating the lack of clarity and consensus.

5.3. *A Call for Common Language and Future Research*

We are not suggesting that scholars should not be innovative, creative and carve their own paths when necessary, but this study underscores the challenges that arise when our collective knowledge and collaboration is unduly fragmented. Without a doubt, this area of study has fundamentally shifted since Danzon-Chambaud's (2021) study, which was published before ChatGPT went public and before the deluge of generative AI tools were available. Danzon-Chambaud (2021) in fact noted that the term "automated journalism" was already contested by some scholars for being too narrow and that it could apply to other algorithmic tasks in journalism, beyond the computer generation of news texts, such as for data-driven investigation or automated fact checking. Now, with the increasing prevalence of not only AI but also generative AI, the underlying understanding of what constitutes automated journalism has fundamentally shifted. While computational and algorithmic in nature, automated news content previously relied on structured data being fed into pre-designed templates that were designed and built in partnership with newsrooms. With the widespread uptake of generative AI tools, which can deal with unstructured data in ways not previously seen, the definitions, theories, concepts, and lines of inquiry around automated news content have ultimately evolved, and along with them, so have the scholarly lines of inquiry.

As such, we ended up with a wildly varied corpus of studies, all of which, indeed, fall under the banner of automated journalism, but which address a broad range of ideas that need to be clarified conceptually. This calls for continued conceptual research into the subject of automated journalism, further dissecting these 185 studies—and inevitably the dozens more that have been completed in the interim—in order to provide the conceptual clarity that is clearly needed and, in doing so, set a foundation for a scholarly consensus in this ever-changing area.

6. Conclusions

In this study, we conducted a systematized review of automated journalism scholarship, with the objective of updating and expanding on a study conducted by Danzon-Chambaud in 2021, which found 33 empirical studies conducted between 2012 and 2020. We include 185 studies in our corpus, 128 of which are fully, 30 tangentially, and 23 partially about automated news content. Within this corpus we included empirical, analytical, applied, and conceptual studies to build a more fulsome picture of the landscape of automated journalism scholarship.

As such, the incursion of generative AI on the journalism domain—and the subsequent reflection of this development in journalism studies research—highlights a key takeaway in our study: scholarly research on the subject matter demonstrates a clear need for conceptual clarity. Researchers use inconsistent and overlapping terms, which not only reflects the speed of AI-driven industrial change, but also the lack of shared academic frameworks to make sense of that change.

Beyond a need for conceptual clarity, another key take-away of our study is that the rapid expansion of AI in journalism research reveals structural blind spots in the field, including a persistent Western bias and the underrepresentation of female scholars, non-English scholarship, and researchers from marginalized communities. Future research should therefore move beyond terminological alignment to actively map and integrate diverse epistemologies, languages, and regional experiences in order to foreground how power, bias, and inequality shape both AI systems and the knowledge produced about them.

In addition to these suggestions, this study sets the scene for further lines of inquiry for future research. Ultimately, the substantial number of studies included within our corpus make it challenging to conduct the nuanced analysis that is deserved. This first calls for a more systematic-style review that includes a quality assessment. This could further include a clearer analysis of the focuses within the study types; an analysis of the mixed studies; a meta-analysis of the experimental studies; an evaluation of critical issues related to the uptake of automated journalism; an exploration into the automated journalism tools in use; a theoretical breakdown and conceptual framework of the terms being used to describe automated journalism; and a deeper look into the scholarship at regional levels.

In terms of theoretical frameworks, scholars would be well advised to borrow from concepts in fields adjacent to or even unrelated to the journalism domain. This may include theories such as the technology acceptance model (TAM) situated in the information systems discipline or the theory of planned behavior (TPB) rooted in the field of social psychology to explain, among other avenues of discovery, why people use and perceive AI-generated content or AI-assisted production differently.

That said, by mapping 185 studies across 12 years, this review provides a solid foundation for the next phase of scholarship—one that must grapple with the disruptive, still-unfolding implications of generative AI for both journalistic practice and norms. As such, it is fair to suggest that the rise of AI and generative AI marks a real watershed moment for automated journalism—one which is broadening the field's scope, scale, and complexity, whilst simultaneously also reshaping its core definitions.

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Appendix A. Full List of Included Articles

Table A1. Full list of studies.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
Study		Title	Author	Country	Year	Journal	Study Type
1	(Ambeth Kumar, 2019)	Efficient daily news platform generation using natural language processing	Ambeth Kumar VD, Thirulokachander VR	India, India	2019	International Journal of Information Technology (Singapore)	Applied
2	(Ayapova, 2022)	AI and Human Created Media Texts: Experiment Results.	Ayapova S M, Skripnikova A I	Kazakhstan, Kazakhstan	2022	Herald of Journalism/Журналистика сериясы	Empirical
3	(Aydin, 2024)	Can Artificial Intelligence Write News: A Research on Determining The Effect of Artificial Intelligence on News Writing Practice.	Aydin B, İnce M	Turkey, Turkey	2024	Intermedia International e-Journal	Empirical
4	(Baptista, 2024)	Are Journalists no Longer Needed? Comparative Analysis of the Perceived Quality of Real News and ChatGPT News	Baptista JP, Gradim A	Portugal, Portugal	2024	Conference Proceedings	Empirical
5	(Barrolleta, 2024)	Artificial intelligence versus journalists: The quality of automated news and bias by authorship using a Turing test	Barrolleta LJ, Sandoval-Martín T	Spain, Spain	2024	Analisi	Empirical
6	(Blankespoor, 2018)	Capital market effects of media synthesis and dissemination: evidence from robo-journalism	Blankespoor E, deHaan E, Zhu C	United States, United States, United States	2018	Review of Accounting Studies	Empirical, Analytical, Applied
7	(Canavilhas, 2022)	Artificial Intelligence and Journalism: Current Situation and Expectations in the Portuguese Sports Media	Canavilhas J	Portugal	2022	Journalism and Media	Empirical
8	(Carlson, 2015)	The Robotic Reporter: Automated journalism and the redefinition of labor, compositional forms, and journalistic authority	Carlson M	United States	2015	Digital Journalism	Empirical, Conceptual

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
Study		Title	Author	Country	Year	Journal	Study Type
9	(Caswell, 2018)	Automated Journalism 2.0: Event-driven narratives: From simple descriptions to real stories	Caswell D, Dörr KN	United Kingdom, Switzerland	2018	Journalism Practice	Applied
10	(Caswell, 2019a)	Structured Journalism and the Semantic Units of News	Caswell D	United Kingdom	2019	Digital Journalism	Conceptual, Applied
11	(Caswell, 2019b)	Automating Complex News Stories by Capturing News Events as Data.	Caswell D, Dörr KN	United Kingdom, Switzerland	2019	Journalism Practice	Conceptual
12	(Cheng, 2024)	SNIL: Generating Sports News From Insights With Large Language Models	Cheng L, Deng D, Xie X, Qiu R, Xu M, Wu Y	China, China, China, China, China, China	2024	IEEE Transactions on Visualization and Computer Graphics	Applied, Empirical
13	(Clerwall, 2014)	Enter the Robot Journalist: Users’ perceptions of automated content	Clerwall C	Sweden	2014	Journalism Practice	Empirical
14	(Cloudy, 2023)	The Str(AI)ght Scoop: Artificial Intelligence Cues Reduce Perceptions of Hostile Media Bias	Cloudy J, Banks J, Bowman ND	United States, United States, United States	2023	Digital Journalism	Empirical
15	(Cools, 2024a)	News Automation and Algorithmic Transparency in the Newsroom: The Case of the Washington Post	Cools H, Koliska M	The Netherlands, United States	2024	Journalism Studies	Empirical
16	(Dangovski, 2021)	We Can Explain Your Research in Layman’s Terms: Towards Automating Science Journalism at Scale	Dangovski R, Shen M, Byrd D, Jing L, Tsvetkova D, Nakov P, Soljacic M	United States, United States, United States, United States, Bulgaria, Qatar, United States	2021	Conference Proceedings	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
	Study	Title	Author	Country	Year	Journal	Study Type
17	(Danzon-Chambaud, 2021)	A systematic review of automated journalism scholarship: Guidelines and suggestions for future research	Danzon-Chambaud S	Ireland	2021	Open Research Europe	Analytical
18	(Danzon-Chambaud, 2023a)	Changing or Reinforcing the “Rules of the Game”: A Field Theory Perspective on the Impacts of Automated Journalism on Media Practitioners	Danzon-Chambaud S, Cornia A	Ireland, Ireland	2023	Journalism Practice	Conceptual
19	(Danzon-Chambaud, 2023b)	Automated news in practice: a cross-national exploratory study	Danzon-Chambaud S	Ireland	2023	Open Research Europe	Analytical, Empirical
20	(Danzon-Chambaud, 2024)	The cultural capital you need to work with automated news: Not only ‘your beautiful piece of work’ but also ‘patterns that emerge’	Danzon-Chambaud S, Cornia A	Ireland, Ireland	2024	Journalism	Empirical
21	(Diakopoulos, 2017)	Algorithmic Transparency in the News Media	Diakopoulos N, Koliska M	United States, United States	2017	Digital Journalism	Empirical
22	(Díaz-Noci, 2020)	Artificial Intelligence Systems-Aided News and Copyright: Assessing Legal Implications for Journalism Practices	Díaz-Noci J	Spain	2020	Future Internet	Analytical
23	(Dierickx, 2018)	Between fear and confidence: The dual relationship between journalists and news automation	Dierickx L	Belgium	2018	Journal of Applied Linguistics and Professional Practice	Analytical
24	(Dierickx, 2020)	The social construction of news automation and the user experience.	Dierickx L	Belgium	2020	Brazilian Journalism Research	Empirical
25	(Dierickx, 2023)	News automation, materialities, and the remix of an editorial process	Dierickx L	Belgium	2023	Journalism	Conceptual
26	(Dörr, 2016)	Mapping the field of Algorithmic Journalism	Dörr KN	Switzerland	2016	Digital Journalism	Empirical, Conceptual

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
	Study	Title	Author	Country	Year	Journal	Study Type
27	(Dörr, 2017)	Ethical Challenges of Algorithmic Journalism	Dörr KN, Hollnbuchner K	Switzerland, Switzerland	2017	Digital Journalism	Conceptual
28	(Duncan, 2024)	Attitudes to automated and human written sport journalism	Duncan S, Kunert J, Karg A	Australia, Germany, Australia	2024	Journalism	Empirical
29	(Ekşioğlu, 2019)	Robot Journalist or Human Journalist?: An Analysis is Over News Articles	Ekşioğlu NBS	Turkey	2019	Conference Proceedings	Empirical
30	(Galily, 2018)	Artificial intelligence and sports journalism: Is it a sweeping change?	Galily Y	Israel	2018	Technology in Society	Conceptual
31	(Gavurova, 2024)	An information-analytical system for assessing the level of automated news content according to the population structure—A platform for media literacy system development	Gavurova B, Skare M, Hynek N, Moravec V, Polishchuk V	Czech, Czech, Czech, Czech, Ukraine	2024	Technological Forecasting and Social Change	Applied, Empirical
32	(Ghosh, 2022)	SpecTextor: End-to-End Attention-based Mechanism for Dense Text Generation in Sports Journalism	Ghosh I, Ivler M, Ramamurthy SR, Roy N	United States, United States, United States, United States	2022	Conference Proceedings	Applied
33	(González-Arias, 2024)	The anthropomorphic pursuit of AI-generated journalistic texts: limits to expressing subjectivity	González-Arias C, Chatzikoumi E, López García	Spain, Chile, Spain	2024	Frontiers in Communication	Empirical
34	(Gotmare, 2024)	A multimodal machine learning approach to generate news articles from geo-tagged images	Gotmare A, Thite G, Bewoor L	India, India, India	2024	International Journal of Electrical and Computer Engineering	Applied
35	(Govindaraju, 2021)	Classifying Fake and Real Neurally Generated News	Govindaraju A, Griffith J	Ireland, Ireland	2021	Conference Proceedings	Applied

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
36 (Graefe, 2018)	Readers' perception of computer-generated news: Credibility, expertise, and readability	Graefe A, Haim M, Haarmann B, Brosius H-B	Germany, Germany, Germany, Germany	2018	Journalism	Empirical
37 (Graefe, 2020)	Automated Journalism: A Meta-Analysis of Readers' Perceptions of Human-Written in Comparison to Automated News.	Graefe A, Bohlken N	Germany, Germany	2020	Media and Communication	Analytical
38 (Haapanen, 2020)	Recycling a genre for news automation: The production of Valtteri the Election Bot	Haapanen L, Leppänen L	Finland, Finland	2020	AILA Review	Applied, Empirical
39 (Haim, 2017)	Automated News: Better than expected?	Haim M, Graefe A	Germany, Germany	2017	Digital Journalism	Empirical
40 (Hamna, 2024)	Artificial Intelligence in the Context of Robot Journalism	Hamna DM, Akbar M, Mau M	Indonesia, Indonesia, Indonesia	2024	Smart Innov. Syst. Technol.	Analytical
41 (Henestrosa, 2023)	Automated journalism: The effects of AI authorship and evaluative information on the perception of a science journalism article	Henestrosa A, Greving H, Kimmerle J	Germany, Germany, Germany	2023	Computers in Human Behavior	Empirical
42 (Henestrosa, 2024a)	Understanding and Perception of Automated Text Generation among the Public: Two Surveys with Representative Samples in Germany.	Henestrosa A, Kimmerle J	Germany, Germany	2024	Behavioral Sciences	Empirical
43 (Henestrosa, 2024b)	The Effects of Assumed AI vs. Human Authorship on the Perception of a GPT-Generated Text	Henestrosa A, Kimmerle J	Germany, Germany	2024	Journalism and Media	Empirical
44 (Hong, 2024)	Can AI Become Walter Cronkite? Testing the Machine Heuristic, the Hostile Media Effect, and Political News Written by Artificial Intelligence	Hong J-W, Chang H-CH, Tewksbury D	South Korea, United States, United States	2024	Digital Journalism	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
Study	Title	Author	Country	Year	Journal	Study Type	
45 (Ioscote, 2024)	Artificial Intelligence in Journalism: A Ten-Year Retrospective of Scientific Articles	Ioscote F, Gonçalves A, Quadros C	Brazil, Portugal, Brazil	2024	Journalism and Media	Analytical, Empirical	
46 (Jamil, 2020)	Artificial Intelligence and Journalistic Practice: The Crossroads of Obstacles and Opportunities for the Pakistani Journalists	Jamil S	United Arab Emirates	2020	Journalism Practice	Empirical	
47 (Jamil, 2023)	Automated Journalism and the Freedom of Media: Understanding Legal and Ethical Implications in Competitive Authoritarian Regime	Jamil S	United Arab Emirates	2023	Journalism Practice	Analytical, Empirical	
48 (Jang, 2022)	Knowledge of automated journalism moderates evaluations of algorithmically generated news	Jang W, Kwak DH, Bucy E	South Korea, United States, United States	2022	New Media and Society	Empirical	
49 (Jang, 2023)	The Effects of Anthropomorphism on How People Evaluate Algorithm-Written News	Jang W, Chun JW, Kim Soojin, Kang YW	South Korea, South Korea, United States, South Korea	2023	Digital Journalism	Empirical	
50 (Jang, 2024)	Knowledge of automated journalism moderates evaluations of algorithmically generated news	Jang W, Kwak DH, Bucy E	South Korea, United States, United States	2024	New Media and Society	Empirical	
51 (Jia, 2020a)	An Eye-Tracking Study of Differences in Reading Between Automated and Human-Written News	Jia C, Gwizdka J	United States, United States	2020	Lecture Notes	Empirical	
52 (Jia, 2020b)	Chinese Automated Journalism: A Comparison Between Expectations and Perceived Quality	Jia C	United States	2020	International Journal of Communication	Empirical	
53 (Jia, 2021a)	Algorithmic or human source? Examining relative hostile media effect with a transformer, Æ-based framework	Jia C, Liu R	United States, United States	2021	Media and Communication	Empirical	

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
54 (Jia, 2021b)	Source Credibility Matters: Does Automated Journalism Inspire Selective Exposure?	Jia C, Johnson TJ	United States, United States	2021	International Journal of Communication	Empirical
55 (Jung, 2017)	Intrusion of software robots into journalism: The public's and journalists' perceptions of news written by algorithms and human journalists	Jung J, Song H, Kim Y, Im H, Oh S	South Korea, South Korea, South Korea, South Korea, South Korea	2017	Computers in Human Behavior	Empirical
56 (Kim, 2016)	Automated news generation for TV program ratings	Kim Soojin, Oh J, Lee J	United States, South Korea, South Korea	2016	Conference Proceedings	Applied, Analytical
57 (Kim, 2017)	Newspaper companies' determinants in adopting robot journalism	Kim Daewon, Kim Soojin	South Korea, United States	2017	Technological Forecasting and Social Change	Empirical, Analytical
58 (Kim, 2018)	Newspaper journalists' attitudes towards robot journalism	Kim Daewon, Kim Seongcheol	South Korea, South Korea	2018	Telematics and Informatics	Empirical
59 (Kim, 2019)	Designing an Algorithm-Driven Text Generation System for Personalized and Interactive News Reading.	Kim Dongwhan, Lee J	South Korea, South Korea	2019	International Journal of Human-Computer Interaction	Empirical
60 (Kim, 2020)	A Decision-Making Model for Adopting AI-Generated News Articles: Preliminary Results	Kim Soojin, Kim B	United States, South Korea	2020	Sustainability	Empirical
61 (Kim, 2021a)	Towards a sustainable news business: Understanding readers' perceptions of algorithm-generated news based on cultural conditioning	Kim Y, Lee H	South Korea, South Korea	2021	Sustainability (Switzerland)	Empirical
62 (Kim, 2021b)	A model for user acceptance of robot journalism: Influence of positive disconfirmation and uncertainty avoidance	Kim Daewon, Kim Suwon	South Korea, South Korea	2021	Technological Forecasting and Social Change	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
	Study	Title	Author	Country	Year	Journal	Study Type
63	(Kothari, 2020)	Challenges for journalism education in the era of automation	Kothari A, Hickerson A	United States, United States	2020	Media Practice and Education	Empirical
64	(Krausová, 2022)	Disappearing Authorship: Ethical Protection of AI-Generated News from the Perspective of Copyright and Other Laws	Krausová A, Moravec V	Czech, Czech	2022	“Journal of Intellectual Property, Information Technology and E-Commerce Law”, Information Technology and E-Commerce Law	Empirical
65	(Kreft, 2023)	(Lost) Pride and Prejudice. Journalistic Identity Negotiation Versus the Automation of Content	Kreft J, Boguszewicz-Kreft M, Fydrych M	Poland, Poland, Poland	2023	Journalism Practice	Empirical
66	(Kuai, 2024)	Unravelling Copyright Dilemma of AI-Generated News and Its Implications for the Institution of Journalism: The Cases of US, EU, and China	Kuai J	Sweden	2024	New Media and Society	Analytical
67	(Kunert, 2020)	Automation in sports reporting: Strategies of data providers, software providers, and media outlets	Kunert J	Germany	2020	Media and Communication	Empirical
68	(Lee, 2017)	Implementation of robot journalism by programming custombot using tokenization and custom tagging	Lee N, Kim K, Yoon T	South Korea, South Korea, South Korea	2017	Conference Proceedings	Empirical
69	(Lee, 2020)	Predicting AI News Credibility: Communicative or Social Capital or Both?	Lee S, Nah S, Chung D, Kim J	South Korea, United States, United States, South Korea	2020	Communication Studies	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
Study		Title	Author	Country	Year	Journal	Study Type
70	(Leppänen, 2017)	Finding and expressing news from structured data	Leppänen L, Munezero M, Sirén-Heikel S, Granroth-Wilding M, Toivonen H	Finland, Finland, Finland, Finland, Finland	2017	Conference Proceedings	Empirical
71	(Leppänen, 2020)	Automated journalism as a source of and a diagnostic device for bias in reporting	Leppänen L, Tuulonen H, Sirén-Heikel S	Finland, Finland, Finland	2020	Media and Communication	Applied
72	(Lewis, 2019a)	Libel by Algorithm? Automated Journalism and the Threat of Legal Liability	Lewis SC, Sanders AK, Carmody C	United States, Qatar, United States	2019	Journalism and Mass Communication Quarterly	Conceptual, Analytical
73	(Lewis, 2019b)	Automation, Journalism, and Human—Machine Communication: Rethinking Roles and Relationships of Humans and Machines in News	Lewis SC, Guzman AL, Schmidt TR	United States, United States, United States	2019	Digital Journalism	Conceptual
74	(Li, 2022)	Technology or content: Which factor is more important in people’s evaluation of artificial intelligence news?	Li Y, Yu M, Li S	China, China, China	2022	Telematics and Informatics	Empirical
75	(Linden, 2017)	Decades of Automation in the Newsroom: Why are there still so many jobs in journalism?	Lindén CG	Finland	2017	Digital Journalism	Empirical
76	(Liu, 2018)	Reading Machine-Written News: Effect of Machine Heuristic and Novelty on Hostile Media Perception	Liu BJ, Wei LW	United States, United States	2018	Conference Proceedings	Empirical
77	(Liu, 2019)	Machine Authorship In Situ: Effect of news organization and news genre on news credibility	Liu BJ	United States	2019	Digital Journalism	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
78 (Melin, 2018)	No landslide for the human journalist—An empirical study of computer-generated election news in Finland	Melin M, Leppänen L, Back A, Södergard C, Munezero M, Toivonen H	Finland, Finland, Finland, Finland, Finland, Finland	2018	IEEE Access	Empirical
79 (L. A. Møller, 2024)	Reinforce, readjust, reclaim: How artificial intelligence impacts journalism's professional claim	Møller LA, Skovsgaard M, de Vreese C	Denmark, Denmark, The Netherlands	2024	Journalism	Conceptual
80 (Montal, 2017)	I, Robot. You, Journalist. Who is the Author?: Authorship, bylines and full disclosure in automated journalism	Montal T, Reich Z	Israel, Israel	2017	Digital Journalism	Empirical
81 (Mooshammer, 2022)	There are (almost) no robots in journalism. An attempt at a differentiated classification and terminology of automation in journalism on the base of the concept of distributed and gradualised action.	Mooshammer S	Germany	2022	Publizistik: Vierteljahreshefte für Kommunikations- forschung	Conceptual
82 (Moran, 2022)	Robots in the News and Newsrooms: Unpacking Meta-Journalistic Discourse on the Use of Artificial Intelligence in Journalism	Moran RE, Shaikh SJ	United States, The Netherlands	2022	Digital Journalism	Analytical
83 (Moravec, 2020)	The robotic reporter in the Czech news agency: Automated journalism and augmentation in the newsroom	Moravec V, Macková V, Sido J, Ekštejn K	Czech, Czech, Czech, Czech	2020	Communication Today	Applied, Empirical
84 (Moravec, 2024)	Human or machine? The perception of artificial intelligence in journalism, its socio-economic conditions, and technological developments toward the digital future	Moravec V, Hynek N, Skare M, Gavurova B, Kubak M	Czech, Czech, Czech, Czech, Slovakia	2024	Technological Forecasting and Social Change	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
85 (Munoriyarwa, 2023)	Artificial Intelligence Practices in Everyday News Production: The Case of South Africa's Mainstream Newsrooms	Munoriyarwa A, Chiumbu S, Motsaathebe G	South Africa, South Africa, South Africa	2023	Journalism Practice	Empirical
86 (Nanekar, 2023)	Automated Journalism Based on Sports Analysis	Nanekar E, Nalawade S, Castelino Z, Rukhande S	India, India, India, India	2023	Conference Proceedings	Applied
87 (Noain-Sánchez, 2022)	Addressing the Impact of Artificial Intelligence on Journalism: the perception of experts, journalists and academics	Noain-Sánchez A	Spain	2022	Communication and Society	Empirical
88 (Oh, 2020)	Understanding User Perception of Automated News Generation System	Oh C, Choi J, Lee S, Park S, Kim Daewon, Song J, Lee J, Suh B, Kim Dongwhan	United States, South Korea, South Korea, South Korea, South Korea, South Korea, South Korea, South Korea	2020	Conference Proceedings	Applied, Empirical
89 (Olsen, 2023)	Enthusiasm and Alienation: How Implementing Automated Journalism Affects the Work Meaningfulness of Three Newsroom Groups	Olsen GR	Norway	2023	Journalism Practice	Empirical
90 (Ombelet, 2016)	Employing Robot Journalists: Legal Implications, Considerations and Recommendations	Ombelet P-J, Kuczerawy A, Valcke P	Belgium, Belgium, Belgium	2016	Conference Proceedings	Analytical
91 (Piasecki, 2024)	AI-generated journalism: Do the transparency provisions in the AI Act give news readers what they hope for?	Piasecki S, Morosoli S, Helberger N, Naudts L	Japan, The Netherlands, The Netherlands, The Netherlands	2024	Internet Policy Review	Empirical, Analytical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
92 (Porlezza, 2022)	The Missing Piece: Ethics and the Ontological Boundaries of Automated Journalism.	Porlezza C, Ferri G	Switzerland, Switzerland	2022	ISOJ Journal	Empirical
93 (Primo, 2015)	Who and what do journalism?: An actor-network perspective	Primo A, Zago G	Brazil, Brazil	2015	Digital Journalism	Analytical, Conceptual
94 (Qin, 2021)	Comparable Study on Readability of Machine Generated News and Human News	Qin Y	China	2021	Conference Proceedings	Empirical
95 (Rossner, 2024)	Do Users Really Care? Evaluating the User Perception of Disclosing AI-Generated Content on Credibility in (Sports) Journalism	Rossner A, Cassel M, Huschens M	Germany, Germany, Germany	2024	Conference Proceedings	Empirical
96 (Sandoval-Martín, 2023)	Research on the quality of automated news in international scientific production: methodologies and results	Sandoval-Martín T, Barrolleta LL	Spain, Spain	2023	Cuadernos Info	Analytical
97 (Sarkar, 2024)	Advancing Cricket Narratives: AI-Enhanced Advanced Journaling in the IPL Using Language Models	Sarkar S, Yashwanth TS, Giri A	India, India, India	2024	Conference Proceedings	Empirical
98 (Schapals, 2020)	Assistance or resistance? Evaluating the intersection of automated journalism and journalistic role conceptions	Schapals AK, Porlezza C	Australia, Switzerland	2020	Media and Communication	Empirical
99 (Schultz, 2017)	Newspaper trust and credibility in the age of robot reporters	Schultz B, Sheffer ML	United States, United States	2017	Journal of Applied Journalism & Media Studies	Empirical
100 (Sirén-Heikel, 2023)	At the crossroads of logics: Automating newswork with artificial intelligence-(Re)defining journalistic logics from the perspective of technologists	Sirén-Heikel S, Kjellman M, Lindén CG	Finland, Finland, Finland	2023	Journal of the Association for Information Science and Technology	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
Study		Title	Author	Country	Year	Journal	Study Type
101	(Tandoc, 2020)	Man vs. Machine? The Impact of Algorithm Authorship on News Credibility	Tandoc EC, Yao LJ, Wu S	Singapore, Singapore, Singapore	2020	Digital Journalism	Empirical
102	(Tandoc, 2022)	What is (automated) news? A content analysis of algorithm-written news articles	Tandoc EC, Wu S, Tan J, Contreras-Yap S	Singapore, Singapore, Singapore, Singapore	2022	Media & Jornalismo	Empirical
103	(Thäsler-Kordonouri, 2023)	Automated Journalism in UK Local Newsrooms: Attitudes, Integration, Impact	Thäsler-Kordonouri S, Barling K	Germany, United Kingdom	2023	Journalism Practice	Empirical
104	(Thäsler-Kordonouri, 2024)a	Too many numbers and worse word choice: Why readers find data-driven news articles produced with automation harder to understand	Thäsler-Kordonouri S, Thurman N, Schwertberger U, Stalph F	Germany, Germany, Germany, Germany	2024	Journalism	Empirical
105	(Thäsler-Kordonouri, 2024)b	What Comes After the Algorithm? An Investigation of Journalists’ Post-editing of Automated News Text	Thäsler-Kordonouri S	Germany	2024	Journalism Practice	Empirical
106	(Thurman, 2017)	When Reporters Get Hands-on with Robo-Writing: Professionals consider automated journalism’s capabilities and consequences	Thurman N, Dörr KN, Kunert J	Germany, Switzerland, Germany	2017	Digital Journalism	Empirical
107	(Toff, 2024)	“Or They Could Just Not Use It?”: The Dilemma of AI Disclosure for Audience Trust in News.	Toff B, Simon FM	United States, United Kingdom	2024	International Journal of Press/Politics	Empirical
108	(Torrijos, 2019)	Automated sports journalism. The AnaFut case study, the bot developed by El Confidencial for writing football match reports	Rojas Torrijos JL, Toural Bran C	Spain, Spain	2019	Doxa Comunicacion	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
109 (Torrijos, 2019)	Automated sports coverage. Case study of bot released by The Washington post during the Río, 2016 and Pyeongchang, 2018 Olympics	Torrijos JLR	Spain	2019	Revista Latina de Comunicacion Social	Empirical
110 (Tosyalı, 2021)	Development of Robot Journalism Application: Tweets of News Content in the Turkish Language Shared by a Bot	Tosyalı H, Aytekin Ç	Turkey, Turkey	2021	Journal of Information Technology Management	Applied
111 (Tsourma, 2021)	An ai-enabled framework for real-time generation of news articles based on big eo data for disaster reporting	Tsourma M, Zamichos A, Efthymiadis E, Drosou A, Tzovaras D	Greece, Greece, Greece, Greece, Greece	2021	Future Internet	Applied
112 (Túñez-López, 2019)	Automation, bots and algorithms in newsmaking. Impact and quality of artificial journalism	Túñez-López J-M, Toural Bran C, Valdiviezo Abad C	Spain, Spain, Spain	2019	Revista Latina de Comunicacion Social	Empirical
113 (Ufarte-Ruiz, 2019)	Algorithms and bots applied to journalism. The case of Narrativa Inteligencia Artificial: Structure, production and informative quality	Ufarte-Ruiz M-J, Manfredi Sánchez JL	Spain, Spain	2019	Doxa Comunicacion	Empirical, Analytical
114 (van Dalen, 2012)	THE ALGORITHMS BEHIND THE HEADLINES How machine-written news redefines the core skills of human journalists	van Dalen A	Denmark	2012	Journalism Practice	Analytical
115 (van Dalen, 2024)	Revisiting the Algorithms Behind the Headlines. How Journalists Respond to Professional Competition of Generative AI	van Dalen A	Denmark	2024	Journalism Practice	Analytical
116 (Waddell, 2018)	A Robot Wrote This?: How perceived machine authorship affects news credibility	Waddell TF	United States	2018	Digital Journalism	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)							
	Study	Title	Author	Country	Year	Journal	Study Type
117	(Waddell, 2019a)	Attribution Practices for the Man-Machine Marriage: How Perceived Human Intervention, Automation Metaphors, and Byline Location Affect the Perceived Bias and Credibility of Purportedly Automated Content	Waddell TF	United States	2019	Journalism Practice	Empirical
118	(Waddell, 2019b)	Can an Algorithm Reduce the Perceived Bias of News? Testing the Effect of Machine Attribution on News Readers’ Evaluations of Bias, Anthropomorphism, and Credibility	Waddell TF	United States	2019	Journalism and Mass Communication Quarterly	Empirical
119	(R. Wang, 2024)	Behind the black box: The moderating role of the machine heuristic on the effect of transparency information about automated journalism on hostile media bias perception	Wang R, Ophir Y	United States, United States	2024	Journalism	Empirical
120	(S. Wang, 2024)	The Impact of Machine Authorship on News Audience Perceptions: A Meta-Analysis of Experimental Studies	Wang S, Huang G	China, Hong Kong–China	2024	Communication Research	Analytical
121	(Weeks, 2014)	Media law and copyright implications of automated journalism.	Weeks L	United States	2014	Journal of Intellectual Property & Entertainment Law	Analytical
122	(Wischnewski, 2022)	Can AI Reduce Motivated Reasoning in News Consumption? Investigating the Role of Attitudes Towards AI and Prior-Opinion in Shaping Trust Perceptions of News	Wischnewski M, Krämer N	Germany, Germany	2022	Frontiers in Artificial Intelligence and Applications	Empirical
123	(Wölker, 2021)	Algorithms in the newsroom? News readers’ perceived credibility and selection of automated journalism	Wölker A, Powell TE	The Netherlands, The Netherlands	2021	Journalism	Empirical

Table A1. Cont.

Studies Tagged as <i>Fully</i> About Automated Journalism (<i>n</i> = 130)						
Study	Title	Author	Country	Year	Journal	Study Type
124 (Wu, 2019a)	When Journalism and Automation Intersect: Assessing the Influence of the Technological Field on Contemporary Newsrooms	Wu S, Tandoc EC, Salmon CT	Singapore, Singapore, Singapore	2019	Journalism Practice	Empirical
125 (Wu, 2019b)	A Field Analysis of Journalism in the Automation Age: Understanding Journalistic Transformations and Struggles Through Structure and Agency	Wu S, Tandoc EC, Salmon CT	Singapore, Singapore, Singapore	2019	Digital Journalism	Empirical
126 (Wu, 2019c)	Journalism Reconfigured: Assessing human—machine relations and the autonomous power of automation in news production	Wu S, Tandoc EC, Salmon CT	Singapore, Singapore, Singapore	2019	Journalism Studies	Empirical
127 (Wu, 2020)	Is Automated Journalistic Writing Less Biased? An Experimental Test of Auto-Written and Human-Written News Stories.	Wu Yanfang	United States	2020	Journalism Practice	Empirical
128 (Young, 2015)	From Mr. and Mrs. Outlier To Central Tendencies: Computational journalism and crime reporting at the Los Angeles Times	Young ML, Hermida A	Canada, Canada	2015	Digital Journalism	Empirical
129 (Zhang, 2023)	Dissecting Automated News Production From a Transdisciplinary Perspective: Methodology, Linguistic Application, and Narrative Genres	Zhang W, Tornero JMP, Tian QS	China, Spain, China	2023	SAGE Open	Analytical, Conceptual
130 (Zheng, 2018)	When algorithms meet journalism: The user perception to automated news in a cross-cultural context	Zheng Y, Zhong B, Yang F	China, China, United States	2018	Computers in Human Behavior	Empirical

Table A1. Cont.

Studies Tagged as <i>Partially</i> About Automated Journalism (<i>n</i> = 25)							
	Study	Title	Author	Country	Year	Journal	Study Type
131	(Albizu-Rivas, 2024)	Artificial Intelligence in Slow Journalism: Journalists’ Uses, Perceptions, and Attitudes	Albizu-Rivas I, Parratt-Fernández S, Parratt-Fernández M	Spain, Spain, Spain	2024	Journalism and Media	Empirical
132	(CalvoRubio, 2024a)	A Methodological Proposal to Evaluate Journalism Texts Created for Depopulated Areas Using AI	Calvo Rubio LM, Ufarte Ruiz MJ, Murcia Verdú FJ	Spain, Spain, Spain	2024	Journalism and Media	Empirical
133	(CalvoRubio, 2024b)	Criteria for journalistic quality in the use of artificial intelligence	Calvo Rubio LM, Rojas Torrijos JL	Spain, Spain	2024	Communication & Society	Empirical
134	(Craig, 2024)	The role of affective and cognitive involvement in the mitigating effects of AI source cues on hostile media bias	Craig MJA, Choi M	United States, South Korea	2024	Telematics and Informatics	Empirical
135	(Fernandes, 2023)	Data Science, Machine learning and big data in Digital Journalism: A survey of state-of-the-art, challenges and opportunities.	Fernandes E, Moro S, Cortez P	Portugal, Portugal, Portugal	2023	Expert Systems with Applications	Analytical
136	(Forja-Pena, 2024a)	The Ethical Revolution: Challenges and Reflections in the Face of the Integration of Artificial Intelligence in Digital Journalism	Forja-Pena T, García-Orosa B, López-García X	Spain, Spain, Spain	2024	Communication & Society	Empirical, Analytical
137	(Heim, 2023)	Consumer Trust in AI-Human News Collaborative Continuum: Preferences and Influencing Factors by News Production Phases	Heim S, Chan-Olmsted S	Germany, United States	2023	Journalism and Media	Empirical
138	(Jia, 2024)	Promises and Perils of Automated Journalism: Algorithms, Experimentation, and “Teachers of Machines” in China and the United States	Jia C, Riedl MJ, Woolley S	United States, United States, United States	2024	Journalism Studies	Empirical

Table A1. Cont.

Studies Tagged as <i>Partially</i> About Automated Journalism (<i>n</i> = 25)						
Study	Title	Author	Country	Year	Journal	Study Type
139 (Kuai, 2022)	AI ≥ Journalism: How the Chinese Copyright Law Protects Tech Giants' AI Innovations and Disrupts the Journalistic Institution	Kuai J, Ferrer-Conill R, Karlsson M	Sweden, Norway, Sweden	2022	Digital Journalism	Empirical
140 (Li, 2024)	Redefining Truth in the Context of AI-Truth Era: A Practice-Led Research of "From Post-Truth to AI-Truth"	Li YL, Chiu CY	Taiwan, Taiwan	2024	Conference Proceedings	Applied
141 (Meier-Vieracker, 2024)	Automated football match reports as models of textuality	Meier-Vieracker S	Germany	2024	Text and Talk	Empirical
142 (H. J. Møller, 2024)	The Algorithmic Gut Feeling—Articulating Journalistic Doxa and Emerging Epistemic Frictions in AI-Driven Data Work	Møller HJ, Thylstrup NB	Denmark, Denmark	2024	Digital Journalism	Empirical
143 (Nah, 2024)	Algorithmic Bias or Algorithmic Reconstruction? A Comparative Analysis Between AI News and Human News	Nah S, Luo J, Kim Seungbae, Chen M, Mitson R, Joo J	United States, United States, United States, United States, United States, United States	2024	International Journal of Communication	Empirical
144 (Peña-Fernández, 2023)	Without journalists, there is no journalism: the social dimension of artificial intelligence in the media	Peña-Fernández S, Meso Ayerdi K, Larrondo-Ureta A, Díaz-Noci J	Spain, Spain, Spain, Spain	2023	Profesional de la Informacion	Analytical
145 (Porlezza, 2024)	The datafication of digital journalism: A history of everlasting challenges between ethical issues and regulation	Porlezza C	Switzerland	2024	Journalism	Empirical, Conceptual
146 (Scheffauer, 2024)	Algorithmic News Versus Non-Algorithmic News: Towards a Principle-based Artificial Intelligence (AI) Theoretical Framework of News Media	Scheffauer R, Gil de Zúñiga H., Correa T	Spain, United States, Chile	2024	Profesional de la información	Conceptual

Table A1. Cont.

Studies Tagged as <i>Partially</i> About Automated Journalism (<i>n</i> = 25)						
Study	Title	Author	Country	Year	Journal	Study Type
147 (Siitonen, 2024)	Mapping Automation in Journalism Studies 2010–2019: A Literature Review	Siitonen M, Laajalahti A, Venäläinen P	Finland, Finland, Finland	2024	Journalism Studies	Analytical
148 (Sonni, 2024)	Bibliometric and Content Analysis of the Scientific Work on Artificial Intelligence in Journalism	Sonni AF, Putri VCC, Irwanto I	Indonesia, Indonesia, Indonesia	2024	Journalism and Media	Analytical
149 (Splendore, 2016)	Quantitatively Oriented Forms of Journalism and Their Epistemology	Splendore S	Italy	2016	Sociology Compass	Conceptual
150 (Strauß, 2019)	Financial journalism in today's high-frequency news and information era	Strauß N	The Netherlands	2019	Journalism	Empirical
151 (Sun, 2022)	Redesigning Copyright Protection in the Era of Artificial Intelligence.	Sun H	Hong Kong–China	2022	Iowa Law Review	Conceptual, Analytical
152 (Tejedor, 2021)	Exo Journalism: A Conceptual Approach to a Hybrid Formula between Journalism and Artificial Intelligence	Tejedor S, Vila P	Spain, Spain	2021	Journalism and Media	Empirical, Analytical, Conceptual
153 (Tessem, 2024)	The future technologies of journalism	Tessem B, Tverberg A, Borch N	Norway, Norway, Norway	2024	Procedia Comput. Sci.	Empirical
154 (Ufarte-Ruiz, 2023)	Use of artificial intelligence in synthetic media: first newsrooms without journalists	Ufarte-Ruiz M-J, Murcia-Verdú F-J, Túñez-López J-M	Spain, Spain, Spain	2023	Profesional de la Informacion	Empirical
155 (Wilczek, 2024)	Transforming the value chain of local journalism with artificial intelligence	Wilczek B, Haim M, Thurman N	Germany, Germany, Germany	2024	AI Magazine	Analytical, Applied

Table A1. Cont.

Studies Tagged as <i>Tangentially</i> About Automated Journalism (<i>n</i> = 30)							
	Study	Title	Author	Country	Year	Journal	Study Type
156	(Bayer, 2024)	Legal implications of using generative AI in the media	Bayer J	Hungary	2024	Information & Communications Technology Law	Conceptual
157	(Bien-Aimé, 2024)	Who Wrote It? News Readers’ Sensemaking of AI/Human Bylines	Bien-Aimé S, Wu M, Appelman A, Jia H	United States, United States, United States, United States	2024	Commun. Rep.	Empirical
158	(Boyles, 2024)	A New(s) Copyright Balancing Act: How American Journalism Institutions Approached the Early Era of Artificial Intelligence and Fair Use	Boyles JL	United States	2024	Journalism Studies	Analytical, Conceptual
159	(Böyük, 2024)	Artificial Intelligence Journalism: An Enquiry within the Framework of News Values and Ethical Principles.	Böyük M	Turkey	2024	Journal of Communication Theory & Research/Iletisim Kuram ve Arastirma Dergisi	Empirical
160	(Breazu, 2024)	ChatGPT-4 as a journalist: Whose perspectives is it reproducing?	Breazu P, Katsos N	United Kingdom, United Kingdom	2024	Discourse & Society	Empirical
161	(CalvoRubio, 2021)	Artificial intelligence and journalism: Systematic review of scientific production in web of science and scopus (2008–2019)	Calvo Rubio LM, Ufarte-Ruiz M-J	Spain, Spain	2021	Communication and Society	Analytical
162	(Carlson, 2018)	Automating judgment? Algorithmic judgment, news knowledge, and journalistic professionalism	Carlson M	United States	2018	New Media and Society	Conceptual
163	(Ceide, 2024)	AI Implementation Strategies in the Spanish Press Media: Organizational Dynamics, Application Flows, Uses and Future Trends	Ceide CF, Vaz-Álvarez M, González IM	Spain, Spain, Spain	2024	Tripodos	Empirical
164	(Cools, 2024b)	News Automation and Algorithmic Transparency in the Newsroom: The Case of the Washington Post.	Cools H, Koliska M	The Netherlands, United States	2024	Journalism Studies	Empirical

Table A1. Cont.

Studies Tagged as <i>Tangentially</i> About Automated Journalism (<i>n</i> = 30)							
	Study	Title	Author	Country	Year	Journal	Study Type
165	(de-Lima-Santos, 2024)	Guiding the way: a comprehensive examination of AI guidelines in global media	de-Lima-Santos M-F, Yeung WN, Dodds T	Australia, United Kingdom, The Netherlands	2024	AI & Society	Empirical
166	(Díaz-Noci, 2024)	The Influence of AI in the Media Workforce: How Companies Use an Array of Legal Remedies.	Díaz-Noci J, Peña-Fernández S, Meso-Ayerdi K, Larrondo-Ureta A	Spain, Spain, Spain, Spain	2024	Tripodos	Analytical
167	(Dierickx, 2024)	A data-centric approach for ethical and trustworthy AI in journalism	Dierickx L, Opdahl AL, Khan SA,L indén CG, Guerrero Rojas DC	Belgium, Norway, Norway, Finland, Norway	2024	Ethics Inf. Technol.	Conceptual
168	(Grimme, 2024)	AI in the newsroom: a collective case study about newsworker-AI collaboration in the German newspaper industry	Grimme M, Zabel C	Germany, Germany	2024	Journal of Media Business Studies	Empirical
169	(Gutiérrez-Caneda, 2024)	Ethics and journalistic challenges in the age of artificial intelligence: talking with professionals and experts	Gutiérrez-Caneda B, Lindén CG, Vázquez-Herrero J	Spain, Finland, Norway	2024	Frontiers in Communication	Empirical
170	(Hermida, 2024)	From automata to algorithms: A jobs-to-be-done approach to AI in journalism	Hermida A	Canada	2024	Estudios sobre el Mensaje Periodístico	Conceptual
171	(Jones, 2019)	Atomising the News: The (In)Flexibility of Structured Journalism	Jones R, Jones B	United Kingdom, United Kingdom	2019	Digital Journalism	Empirical
172	(Klimashevskaja, 2021)	Automatic News Article Generation from Legislative Proceedings: A Phenom-Based Approach	Klimashevskaja A, Gadgil R, Gerrity T, Khosmood F, Gütl C, Howe P	Austria, United States, United States, United States, Austria, United States	2021	Lecture Notes	Applied, Empirical

Table A1. Cont.

Studies Tagged as <i>Tangentially</i> About Automated Journalism (<i>n</i> = 30)							
Study	Title	Author	Country	Year	Journal	Study Type	
173	(Komatsu, 2020)	AI should embody our values: Investigating journalistic values to inform AI technology design	Komatsu T, Lopez MG, Makri S, Porlezza C, Cooper G, MacFarlane A, Missaoui S	United Kingdom, United Kingdom, United Kingdom, Switzerland, United Kingdom, United Kingdom, United Kingdom	2020	Conference Proceedings	Empirical
174	(Mahony, 2024)	Concerns about the role of artificial intelligence in journalism, and media manipulation	Mahony S, Chen Q	United Kingdom, China	2024	Journalism	Conceptual
175	(Milosavljević, 2019)	Human Still in the Loop: Editors Reconsider the Ideals of Professional Journalism Through Automation	Milosavljević M, Vobič I	Slovenia, Slovenia	2019	Digital Journalism	Empirical
176	(Milosavljević, 2021)	“Our task is to demystify fears”: Analysing newsroom management of automation in journalism	Milosavljević M, Vobič I	Slovenia, Slovenia	2021	Journalism	Empirical
177	(L. A. Møller, 2025)	A Little of that Human Touch: How Regular Journalists Redefine Their Expertise in the Face of Artificial Intelligence	Møller LA, van Dalen A, Skovsgaard M	Denmark, Denmark, Denmark	2025	Journalism Studies	Empirical
178	(Montaña-Niño, 2024)	Beyond the “critical incident”: COVID-19, data journalism and the slow road to editorial automation in Australian newsrooms	Montaña-Niño SX, Burgess J	Australia, Australia	2024	New Media and Society	Empirical
179	(Nocera, 2021)	Crosstown Foundry: A Scalable Data-driven Journalism Platform for Hyper-local News	Nocera L, Constantinou G, Tran LV, Kim Seon Ho, Kahn G, Shahabi C	United States, United States, United States, United States, United States, United States	2021	Conference Proceedings	Empirical

Table A1. Cont.

Studies Tagged as <i>Tangentially</i> About Automated Journalism (<i>n</i> = 30)						
Study	Title	Author	Country	Year	Journal	Study Type
180 (Owsley, 2024)	Awareness and perception of artificial intelligence operationalized integration in news media industry and society	Owsley CS, Greenwood K	United States, United States	2024	AI & Society	Empirical
181 (Quinonez, 2024)	A new era of AI-assisted journalism at Bloomberg	Quinonez C, Meij E	United Kingdom, United Kingdom	2024	AI Magazine	Analytical, Conceptual
182 (Shilina, 2023)	Artificial journalism: the reverse of human-machine communication paradigm. Mapping the field of AI critical media studies	Shilina MG, Volkova II, Bombin AYu, Smirnova AA	Russia, Russia, Russia, Russia	2023	RUDN Journal of Studies in Literature and Journalism	Conceptual
183 (Sigsgaard, 2024)	Striking the (im)balance: a review of the relative prevalence of meta-ethical models in AI journalism research	Sigsgaard ME	Denmark	2024	Journalism	Analytical
184 (Stalph, 2024)	Exploring audience perceptions of, and preferences for, data-driven “quantitative” journalism	Stalph F, Thurman N, Thäsler-Kordonouri S	Germany, Germany, Germany	2024	Journalism	Empirical
185 (Yu, 2021)	Friend or foe? Human journalists’ perspectives on artificial intelligence in Chinese media outlets.	Yu Yang, Huang Kuo	China, China	2021	Chinese Journal of Communication	Empirical

Appendix B. PRISMA Checklist

Section and Topic	Item #	Checklist Item	Location Where Item Is Reported
Title			
Title	1	Identify the report as a systematic review.	The paper is identified as a systematized review in the title.
Abstract			
Abstract	2	See the PRISMA 2020 for Abstracts checklist (Table 2).	n/a—this is a full paper
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 5, in the Section 2.4. <i>Systematically Reviewing Automated Journalism Scholarship</i>
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 6, summarized in the research questions.
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Inclusion criteria are summarized on page 7 in the Section 3.1.1. <i>Inclusion Criteria</i> . Exclusion criteria are listed on page 8 under Section 3.2. <i>Data Collection</i> .
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Databases are listed beginning on page 7 in the Section 3.1.3. <i>Databases</i> . Collection dates are included on page 8 under Section 3.2. <i>Data Collection</i> .
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 7, in the Section 3.1. <i>Search Strategy</i> .
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 8 beginning in the Section 3.2. <i>Data Collection</i> and continued in the Section 3.3. <i>Data Analysis</i> .
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 8 beginning in the Section 3.2. <i>Data Collection</i> and continued in the Section 3.3. <i>Data Analysis</i> .

Section and Topic	Item #	Checklist Item	Location Where Item Is Reported
Methods			
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7 in the Section 3.1.1. <i>Inclusion Criteria</i> .
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 9 in the Section 3.3. <i>Data Analysis</i> .
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	n/a—This systematized review does not include a quality assessment of the included studies.
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	n/a—This systematized review does not include a quality assessment of the included studies.
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 7 in the Section 3.1.1. <i>Inclusion Criteria</i> and page 8 under Section 3.2. <i>Data Collection</i> .
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Any synthesis methods or calculations are reported alongside specific results, beginning on Page 9.
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results.	

Section and Topic	Item #	Checklist Item	Location Where Item Is Reported
Methods			
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	n/a—This systematized review does not include a quality assessment of the included studies.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	n/a—This systematized review does not assess health outcomes.
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (see Figure 1).	Page 8 beginning in the Section 3.2. <i>Data Collection</i> and continued in the Section 3.3. <i>Data Analysis</i> .
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 8 under Section 3.2. <i>Data Collection</i> describes studies from the original corpus that were excluded.
Study characteristics	17	Cite each included study and present its characteristics.	Appendix A
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.	n/a—This systematized review does not include a quality assessment of the included studies, does not address health issues and does not include any statistical synthesis.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	

Section and Topic	Item #	Checklist Item	Location Where Item Is Reported
Discussion			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 18–21 in Section 5. <i>Discussion</i>
	23b	Discuss any limitations of the evidence included in the review.	n/a—This systematized review does not include a quality assessment of evidence from the included studies.
	23c	Discuss any limitations of the review processes used.	Issues with inconsistent terms is discussed throughout, but addressed specifically on Pages 20–21 in Section 5. <i>Discussion</i> .
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 18–21 in Section 5. <i>Discussion</i> and pages 21–22 in Section 6. <i>Conclusion</i> .
Other information			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	n/a—This systematized review does not assess health or medical issues, does not include a quality assessment of the included studies, and is not registered.
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 22
Competing interests	26	Declare any competing interests of review authors.	Page 22
Availability of data, code, and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data are available from authors on request.

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